ZOO 1302
Vertebrate Diversity

Dr. (Mrs.) T.V. Sundarabarathy
Department of Biology
Faculty of Applied Sciences
Rajarata University of Sri Lanka

Superclass -Gnathostomata

Cartilage fish

&

Bony fish

Learning Outcomes

At the end of the lesson you would be able to describe ……

- the general characters of superclass Gnathostomata.
- ancestry and relationships of major groups of fishes.
- the characters of living fish classes.

Phylum : Chordata

Subphylum : Vertebrata

Superclass : Gnathostomata

Class : Placodermi (armored gnathostomes) *extinct*

Class : Chondrichthyes (cartilaginous fish)

Class : Actinopterygii (ray-finned fish)

Class : Sarcopterygii (lobe-finned fish)

Class : Amphibia(amphibians)

Class : Reptilia (reptiles)

Class : Aves(birds)

Class : Mammalia(mammals)

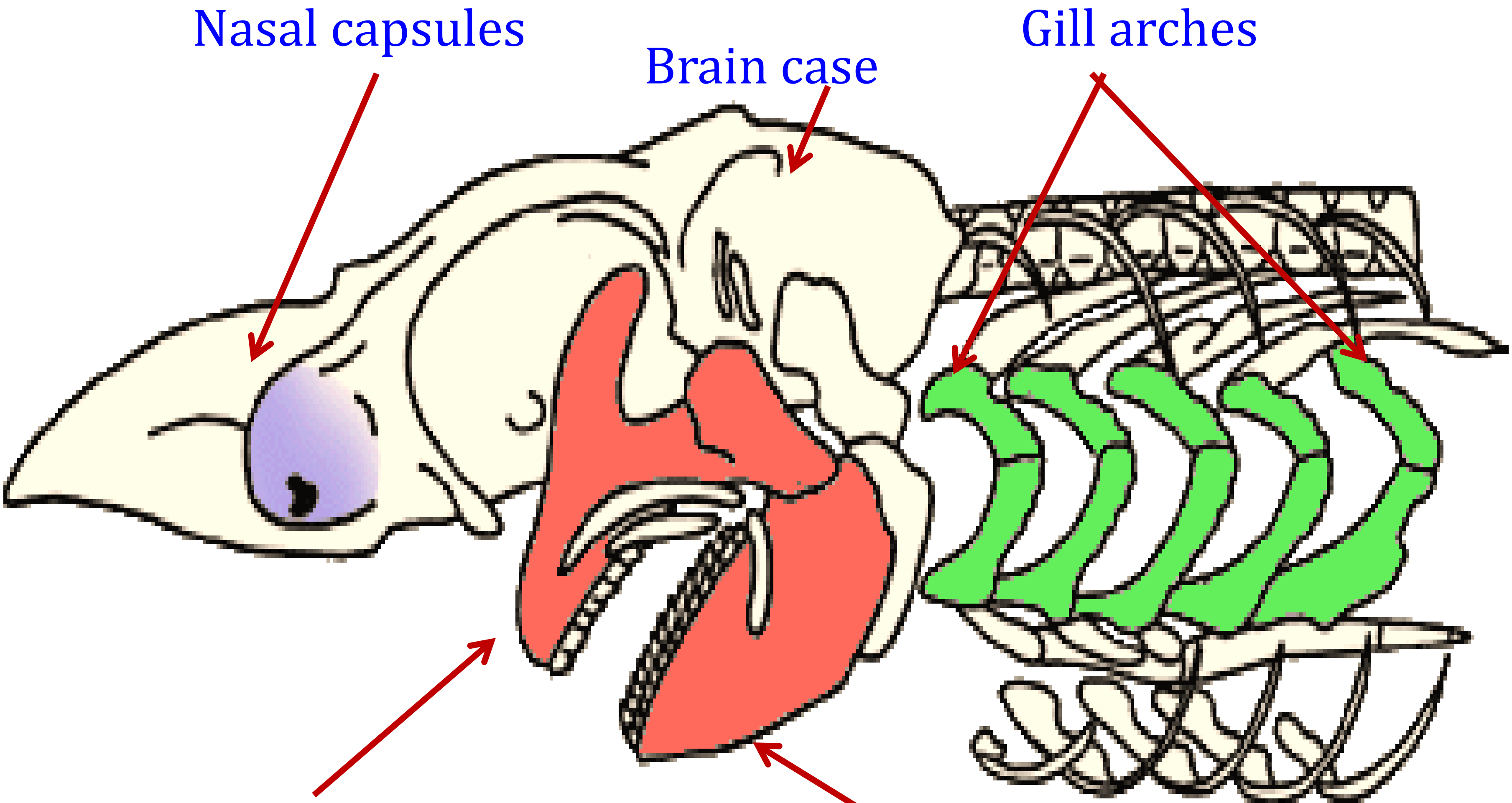
Superclass : Gnathostomata

- a vertebrate animal possessing true jaws
- includes jawed fishes and tetrapods
- evolved 400 million years ago

Characteristic features of Gnathostomata

1. Vertically biting device - jaws

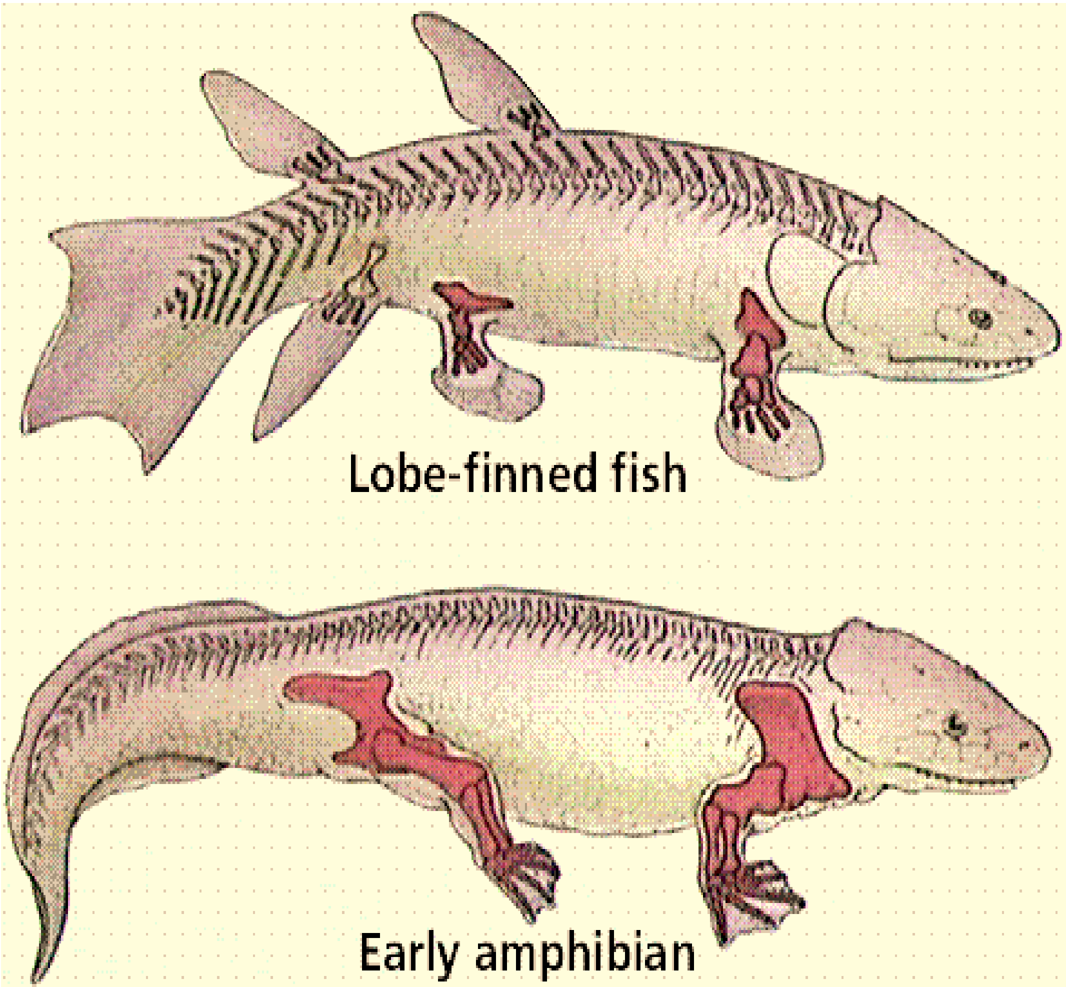
- primitively made up by two endoskeletal elements
    - palatoquadrate cartilage
    - Meckel's cartilage
- Dermal elements
    - teeth, sometimes attached to large dermal bones.



Skull of a gnathostome, or jawed vertebrates (eg. shark)

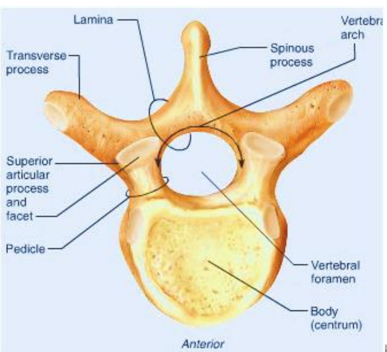
2. Paired appendages

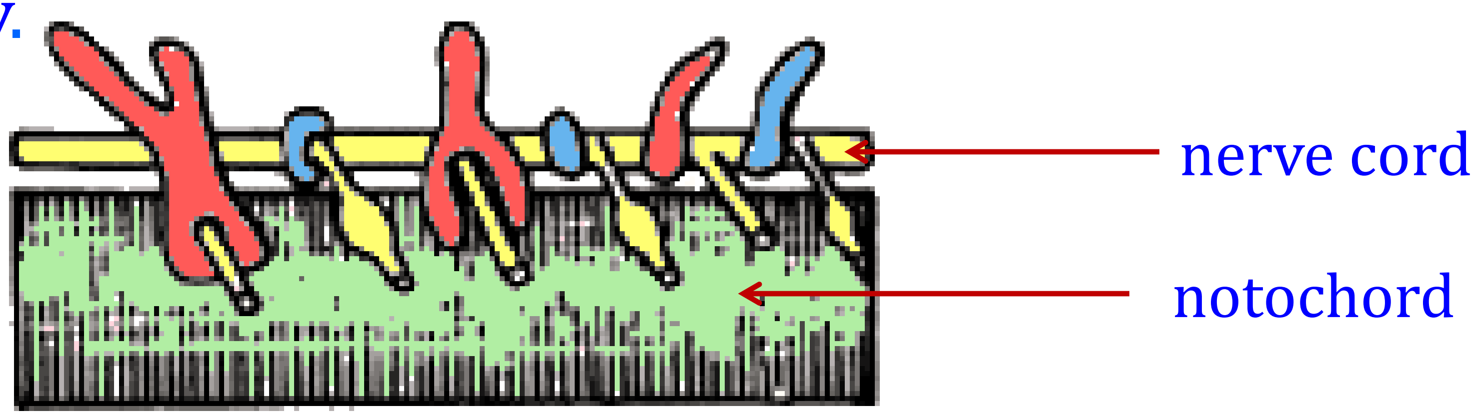
- presence of paired pectoral & pelvic appendages in the form of fins or limbs



3. Interventrals and basiventrals in the vertebral column

- These are the elements of the backbone which lie under the notochord, and match the basidorsals and interdorsals respectively.

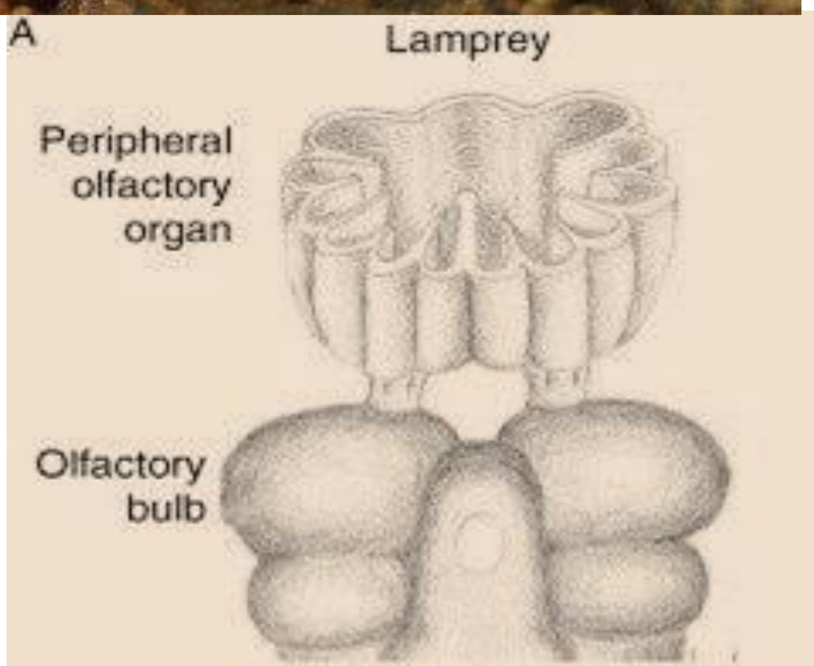


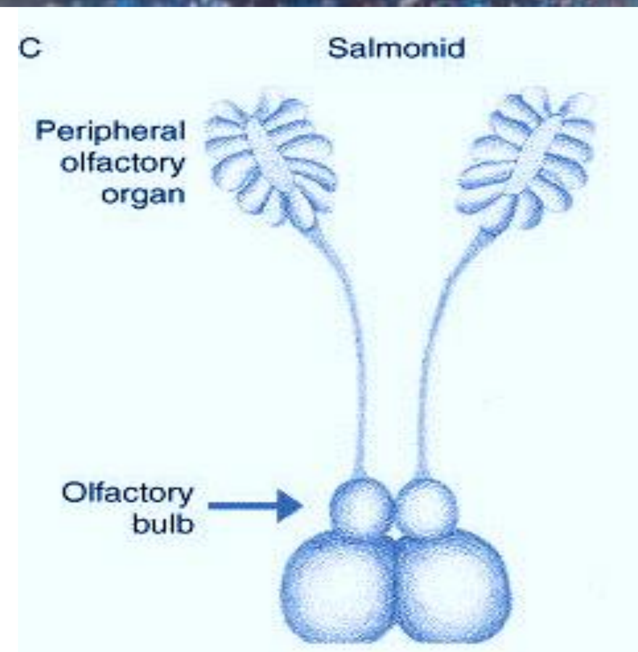


vertebral column of lamprey

vertebral column of gnathostomata

4. Paired nasal sacs are independent from the hypophysial tube



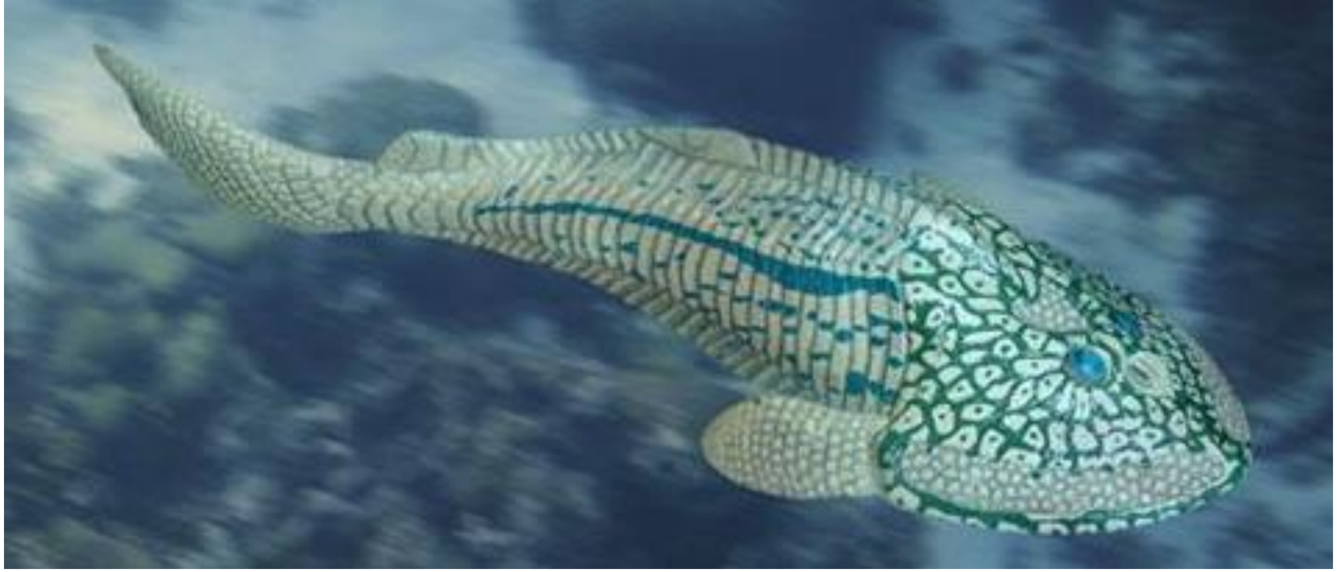


Ancestry and Relationships of Major Groups of Fishes

- The earliest fishlike vertebrates were a paraphyletic assemblage *(descended from a common evolutionary ancestor or ancestral group, but not including all the descendant groups)* of jawless agnathan fishes, the ostracoderms.

- One group of the ostracoderms gave rise to the jawed gnathostomes.

- The jawless agnathans, the least derived of the two groups, include along with the extinct ostracoderms the living hagfishes and lampreys.

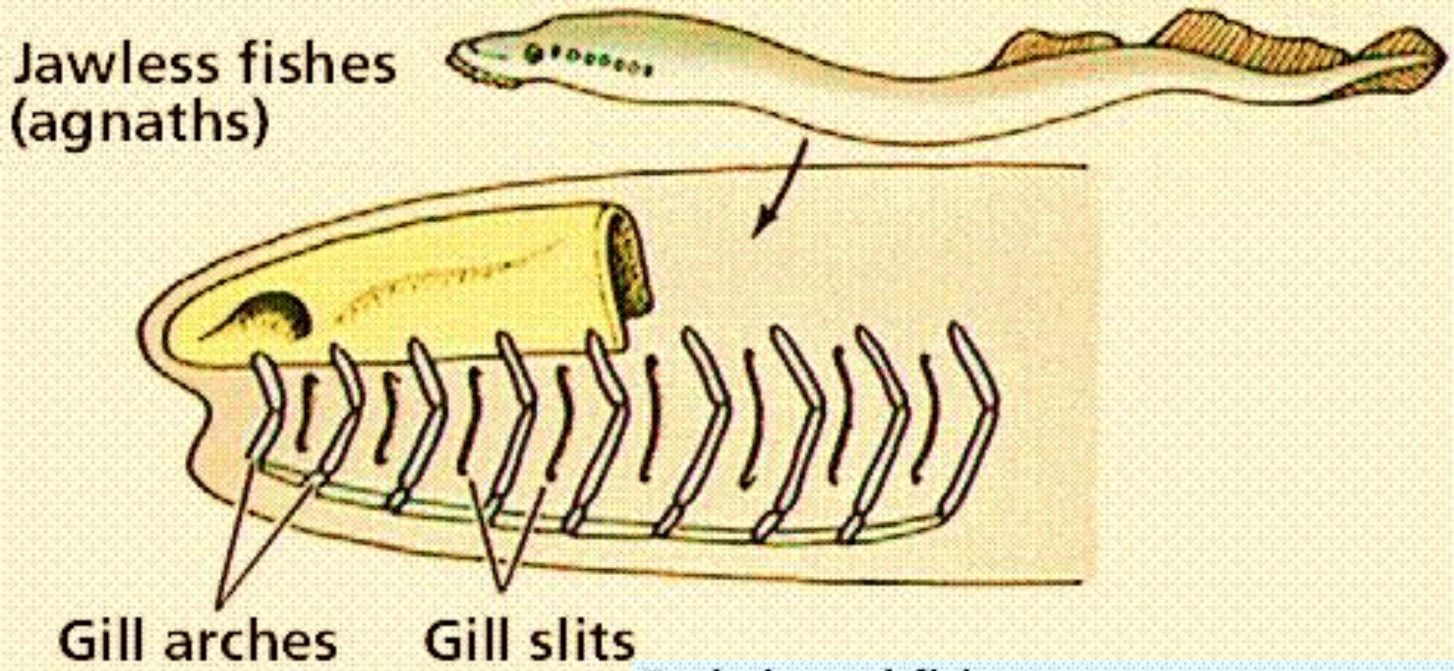


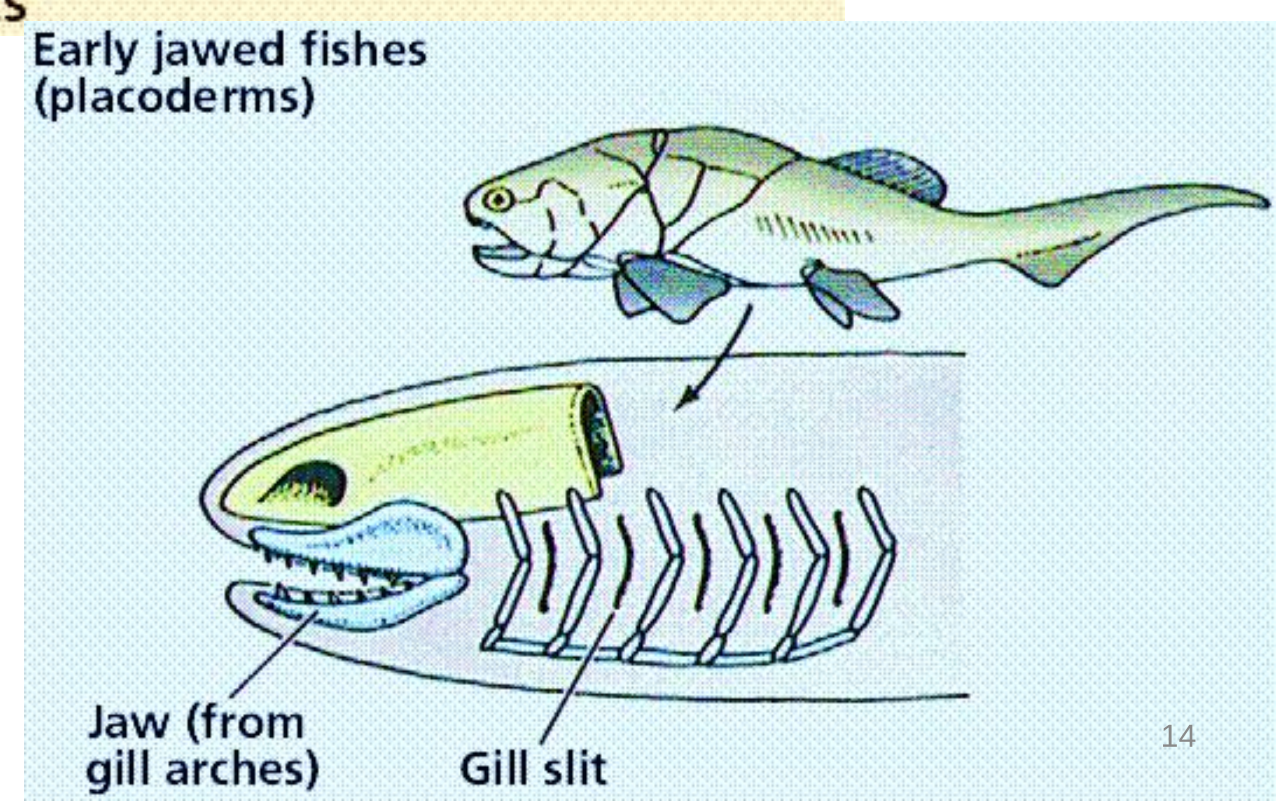
Ostracoderm

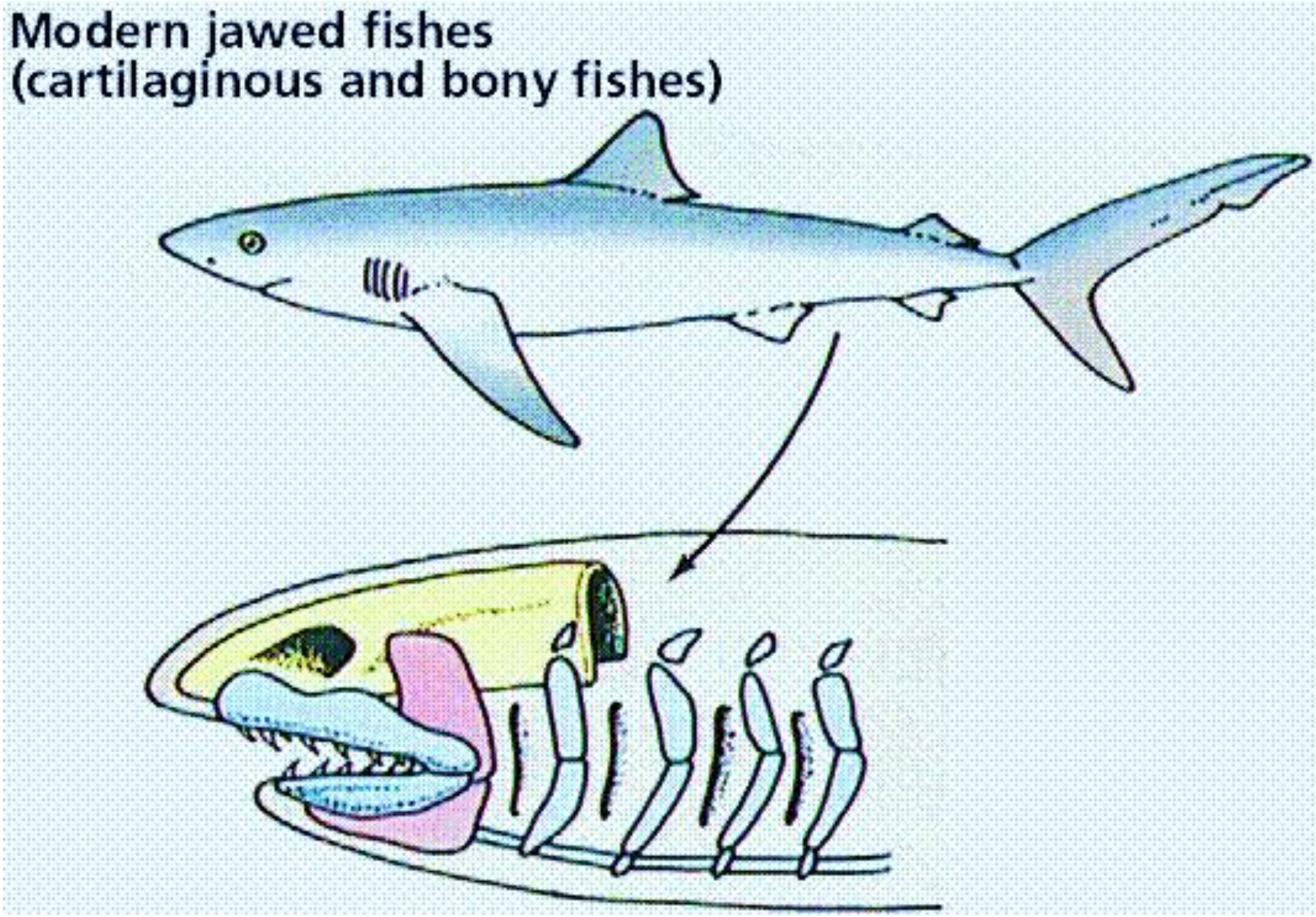
Hagfishes

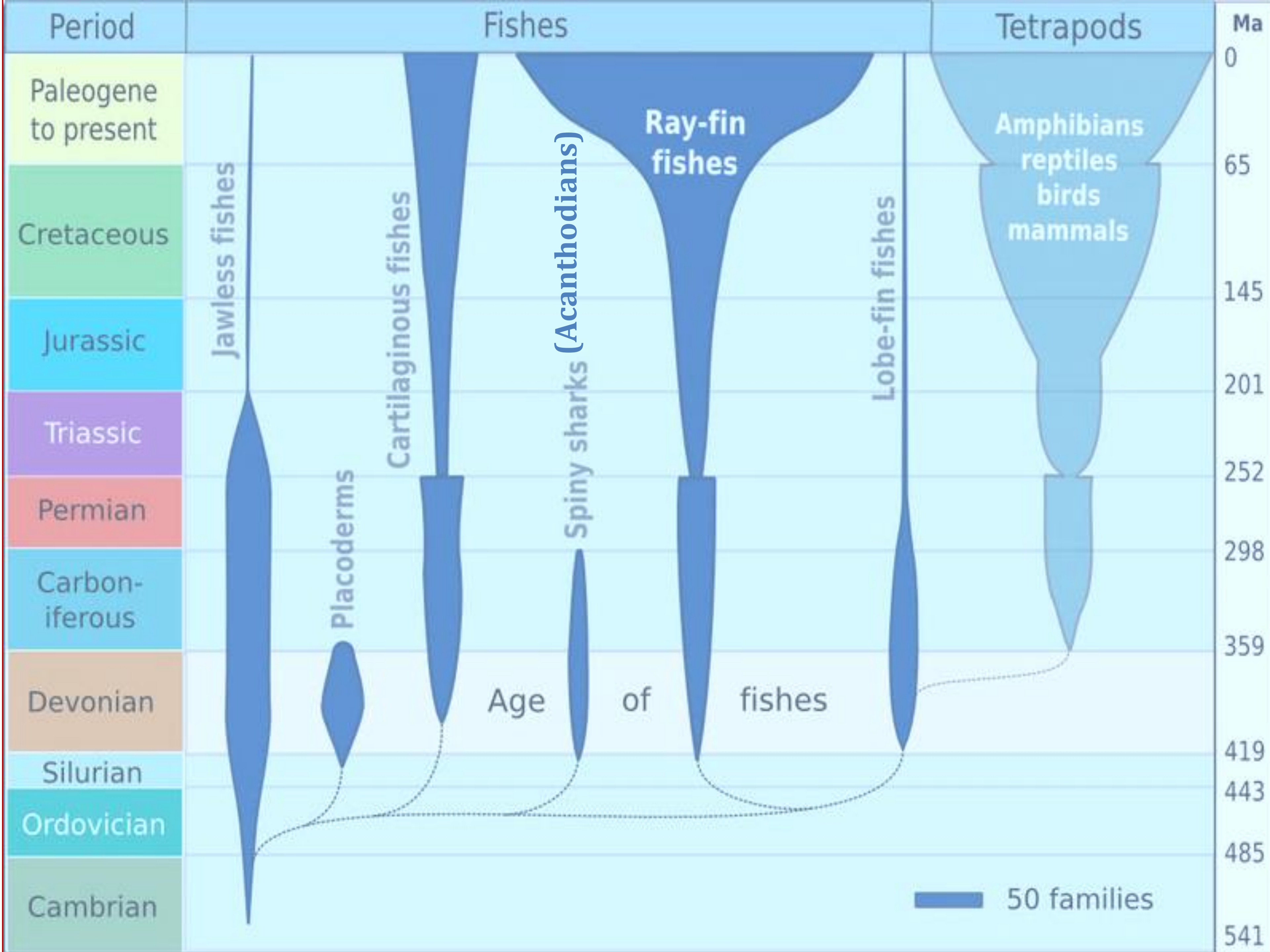
Lampreys

- All remaining fishes and tetrapods (land vertebrates) have
  - ✓ paired appendages and
  - ✓ jaws
- They are in the monophyletic lineage of gnathostomes.
- They appear in the fossil record in the - late Silurian period
- They are with fully formed jaws.
- No forms intermediate between agnathans and gnathostomes.


Jawless fishes
(agnaths)
Gill arches
Gill slits


Early jawed fishes
(placoderms)
Jaw (from
gill arches)
Gill slit


Modern jawed fishes
(cartilaginous and bony fishes)

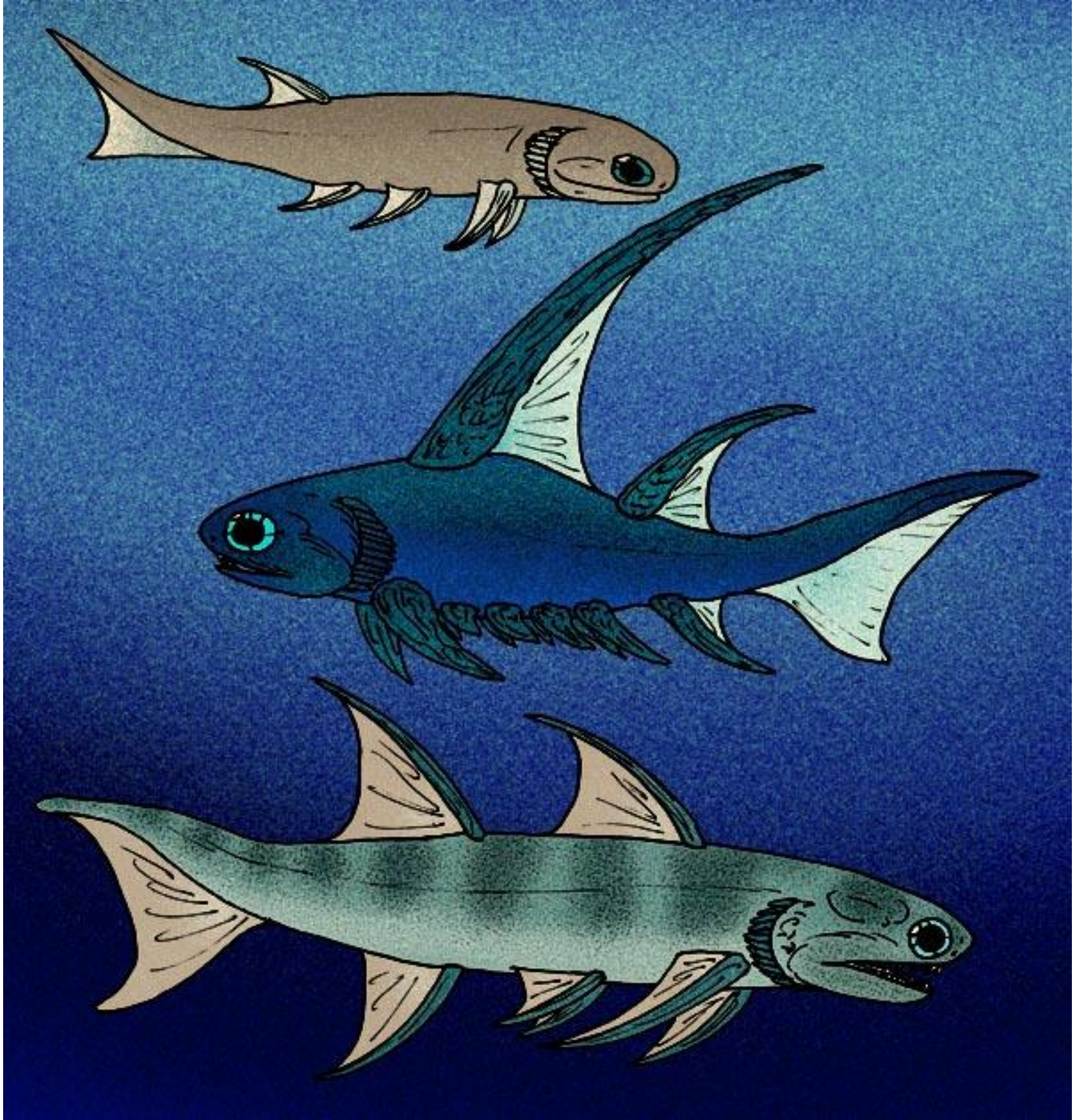

Period
Fishes
Tetrapods
Ma
Paleogene to present
Cretaceous
Jurassic
Triassic
Permian
Carbon-iferous
Devonian
Silurian
Ordovician
Cambrian
0
65
145
201
252
298
359
419
443
485
541
Jawless fishes
Placoderms
Cartilaginous fishes
Spiny sharks (Acanthodians)
Ray-fin fishes
Lobe-fin fishes
Amphibians reptiles birds mammals
Age of fishes
50 families

- The Age of Fishes - Devonian period
- Several distinct groups of jawed fishes were well represented from Devonian period.
- Placoderms - *first group* - they became extinct in the following Carboniferous period, leaving no direct descendants.
- Cartilaginous fishes - *Second group* - the class Chondrichthyes (sharks, rays, and chimaeras),
  - ✓ lost the heavy dermal armor of early jawed fishes
  - ✓ adopted cartilage rather than bone for the skeleton.
  - ✓ most are active predators with shark like or ray like body forms that have undergone only minor changes over the ages.

- Devonian and Carboniferous periods of the Paleozoic era - A group, sharks and their kin flourished
  - ✓ but declined dangerously close to extinction at the end of the Paleozoic.
  - ✓ They staged a recovery in the early Mesozoic and radiated to form the modest but thoroughly successful assemblage of modern sharks and rays

- In Devonian period - the other two groups of gnathostome fishes,
  - ✓ acanthodians and
  - ✓ bony fishes, were well represented.

- Acanthodians (spiny sharks) - somewhat resembled bony fishes but were distinguished by having heavy spines on all fins except the caudal fin.
  - ✓ They became extinct in the lower Permian period.
  - ✓ Although the affinities of the acanthodians are much debated, many authors believe that they are the sister group of the bony fishes.



- Fish come in all shapes and sizes and can be found in a range of habitats.

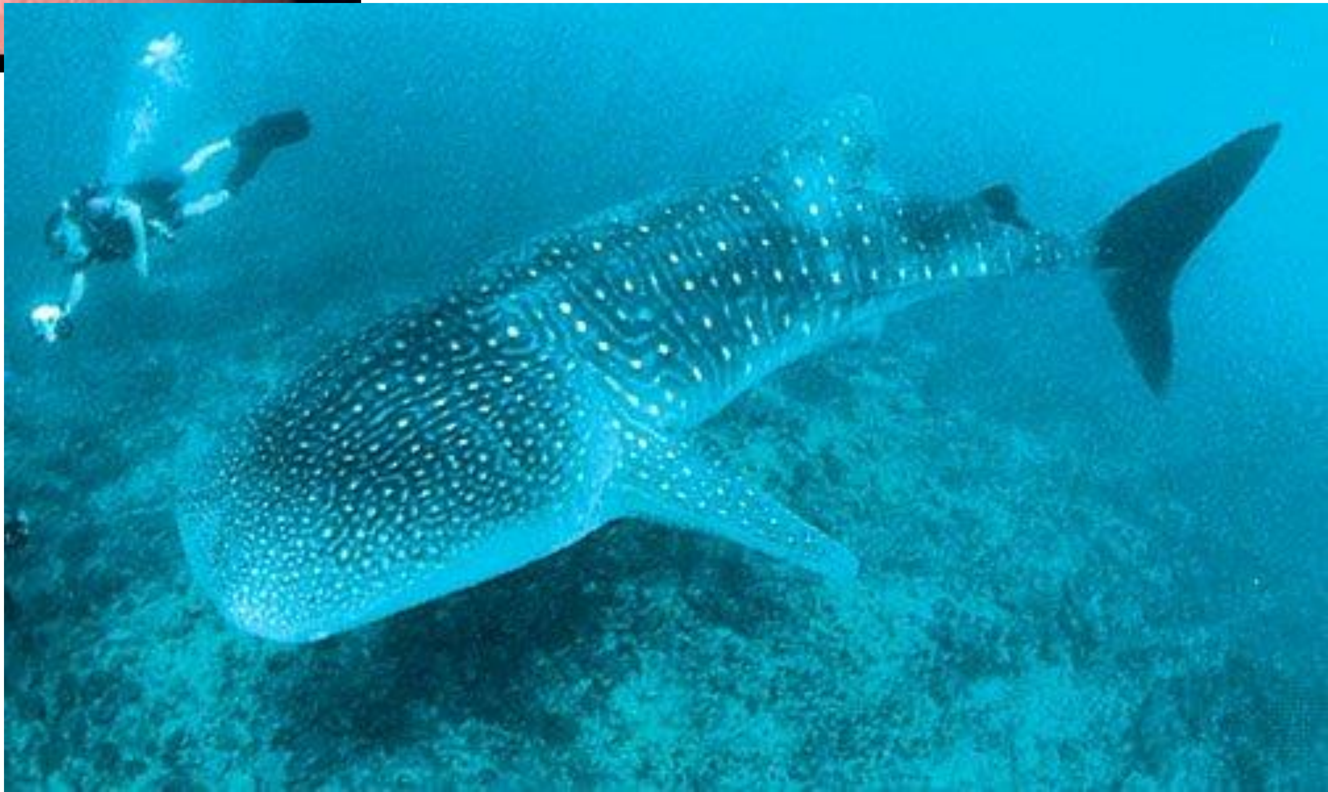
- Fishes also range in size

  ✓ from very small, tiny cyprinid

  - *Paedocypris progenetica* - maximum length of 10 mm

  ✓ to large whale shark *Rhincodon typus* - reaches 12m

Paedocypris progenetica



Rhincodon typus

- Fishes live down deep and up high—from the deepest depths of the oceans to mountain pools high in the Andes.
- Fishes live in warm and cold water—from desert pools to under Arctic pack ice.
- Fishes have a wide range of diets, including plankton, algae, fish, seals, and turtles.
- Some fishes are parasites, some are blind, some are venomous, and some can even produce electricity.
- With all of this amazing diversity, it is challenging to make a definition that describes all fishes.

Definition of Fish

- Defining a fish can also be difficult because the word *fish* is often used to describe things that are not fish.
- Eg. starfish are really echinoderms. Jellyfish are cnidarians., Crayfish are crustaceans

- Even though starfish, jellyfish, and crayfish live in the water, they do not have backbones, and they are not true fish.

- In *Exploring Our Fluid Earth*, starfish are called sea stars and jellyfish are called jelly medusa to avoid confusion with true fish.

- *There are two generally accepted scientific definitions of a fish:*
- Fishes are "*aquatic vertebrate(s) with gills and with limbs in the shape of fins*." (Written by Peter Nelson in 1994.)
- Fishes are "*poikilothermic, aquatic chordate(s) with appendages (when present) developed as fins, whose chief respiratory organs are gills and whose body is usually covered with scales.*" (Written by Tim Berra in 1981.)
- Notice that neither of these definitions mention skeletal features.
- Both definitions include fishes with skeletons made of bone (like tuna) and fishes with skeletons made of cartilage (like sharks).


General Characters of Fish


General Characters of Fishes

1. Cold-blooded aquatic vertebrates.
2. Adapted to fresh or marine water.
3. Body shape - usually streamlined but some are elongated snake-like and few are dorsoventrally compressed.
4. Skin - covered with scales, dermal denticles or bony plates.
5. Tail - highly muscular and helps in propulsion through water.
6. Cranium - completely encloses the brain.

7. Jaws - hinged.

8. Paired limbs - not evolved.

9. Fins - pectoral and pelvic fins are usually present.

- In addition unpaired fins are also present.

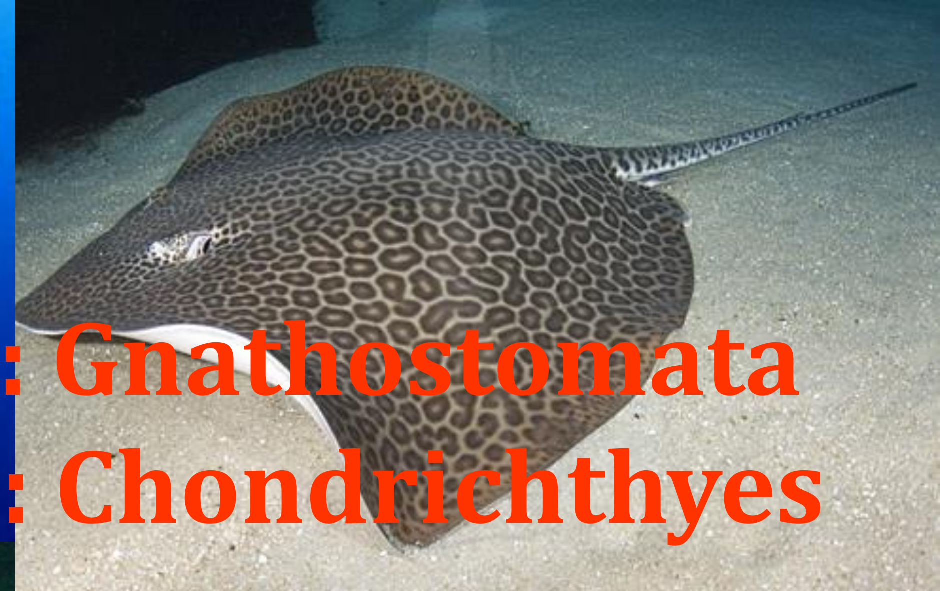
9. Endoskeleton - cartilaginous or bony.

10. Nostrils - paired but do not open into pharynx, except in lung fishes and lobbed fishes.

11. Tactile organs - in the form of barbels are present in some bony fishes.

Super class : Gnathostomata

Class : Chondrichthyes


- Cartilaginous Fish


Cartilaginous Fishes

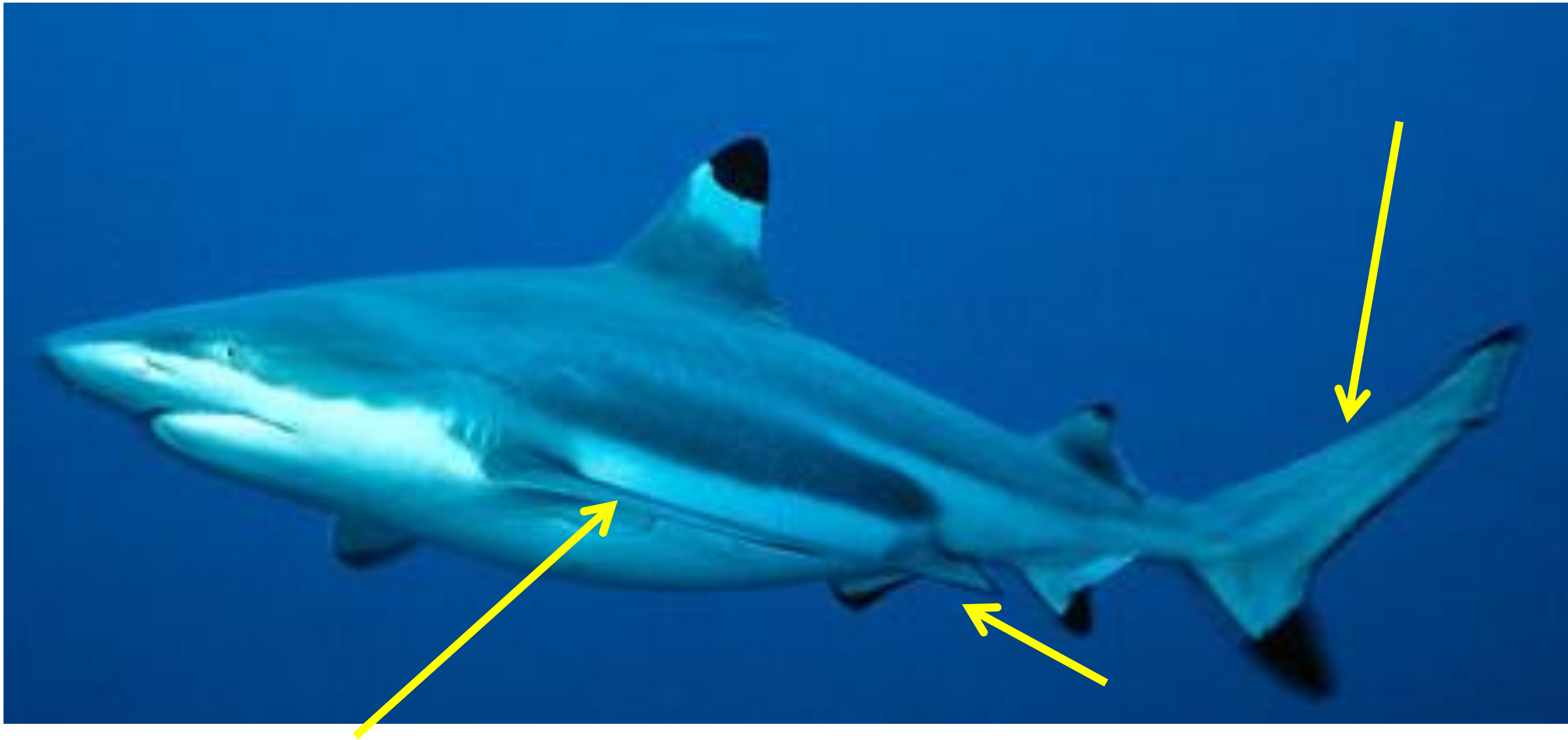
- Nearly 850 species living today
- An ancient, compact, and highly developed group.
- Although a much smaller and less diverse assemblage than the bony fishes their impressive combination of
  - ✓ well-developed sense organs,
  - ✓ powerful jaws ,
  - ✓ swimming musculature, and
  - ✓ predaceous habits

ensures them a secure and lasting place in the aquatic community.

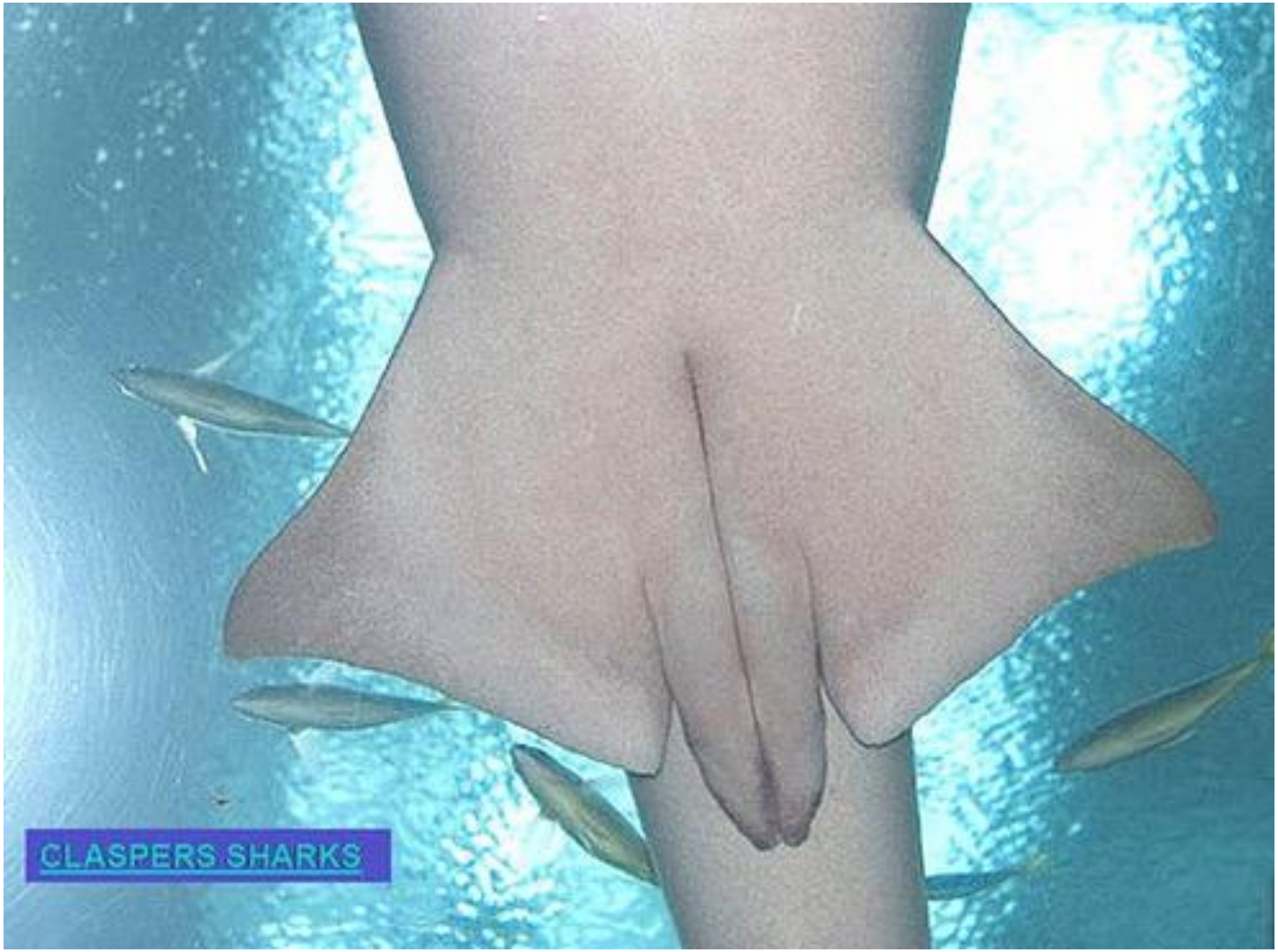
- Most are marine –only 28 species live primarily in freshwater.
- Skeleton - exclusively cartilaginous
- With the exception of whales, sharks include the largest living vertebrates.
- The larger sharks may reach 12 m in length.
- Dogfish sharks so widely studied in zoological laboratories rarely exceed 1 m.

- Large body fusiform or dorsoventrally depressed

- Heterocercal caudal fin & paired pectoral and pelvic fins



- Pelvic fins in males - modified as claspers which is used in copulation



- Teeth - not fused to jaws and usually replaced
- Intestine - with spiral valve
- Fertilization - all chondricthyans shows internal fertilization
- Maternal support of the embryo - highly variable oviparous, ovoviviparous, viviparous

Two subclasses

a) Elasmobranchii - sharks, rays , skates & sawfish

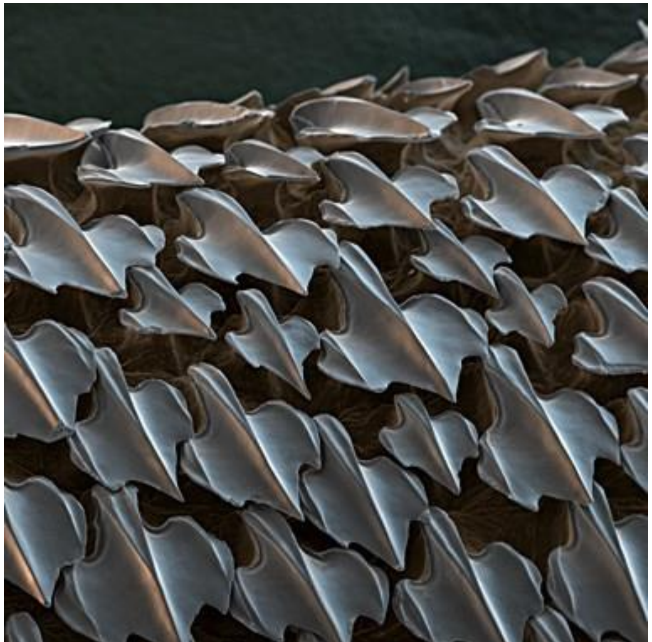
b) Holocephali - chimaeras

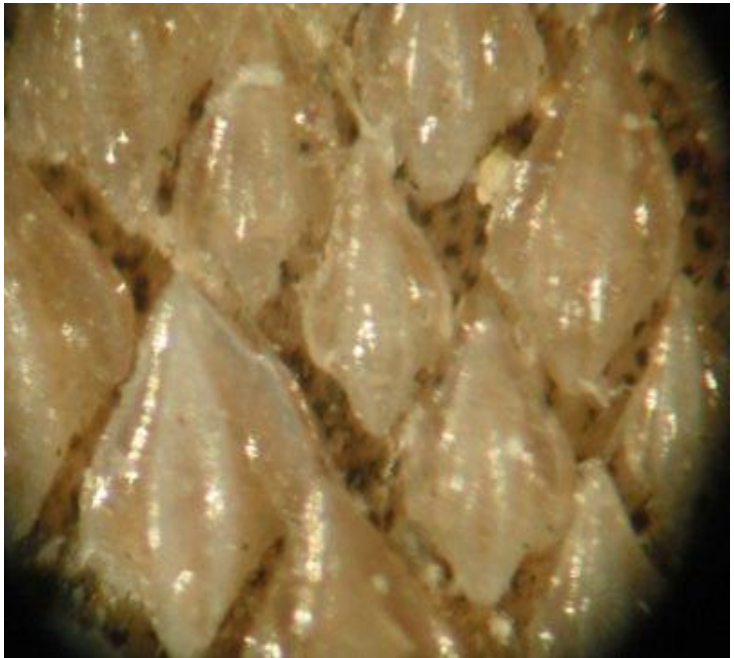
Subclass – Elasmobranchii

- There are nine living orders of elasmobranchs, numbering about 815 species.

Characteristics

- Skin - with placoid scales or derivatives
  - arranged to reduce the turbulence of water flowing along the body surface during swimming.





Placoid scales consist of ……

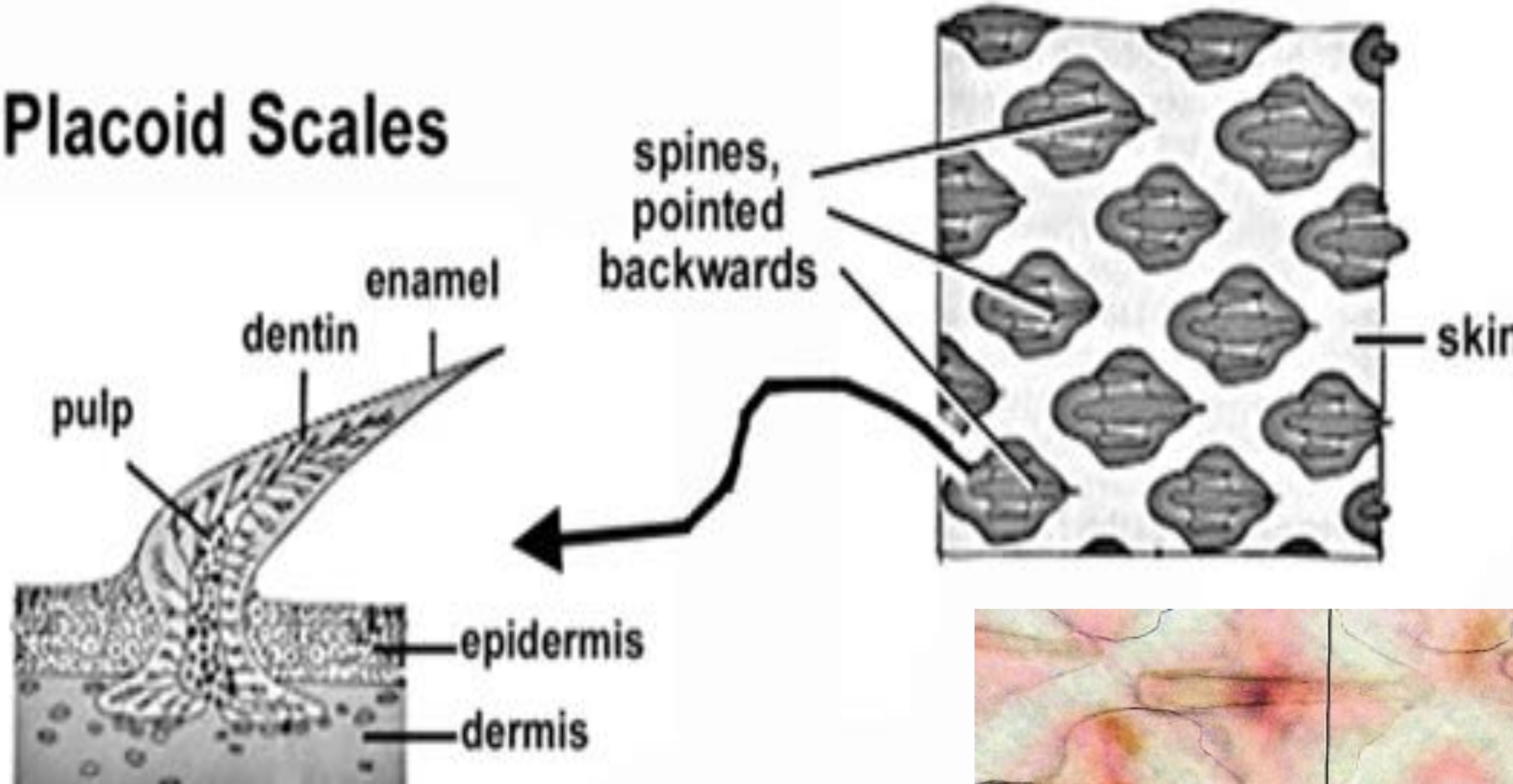
- a basal plate - embedded in the dermis with a caudally directed spine , project through the epidermis
- Basal plate
- spine

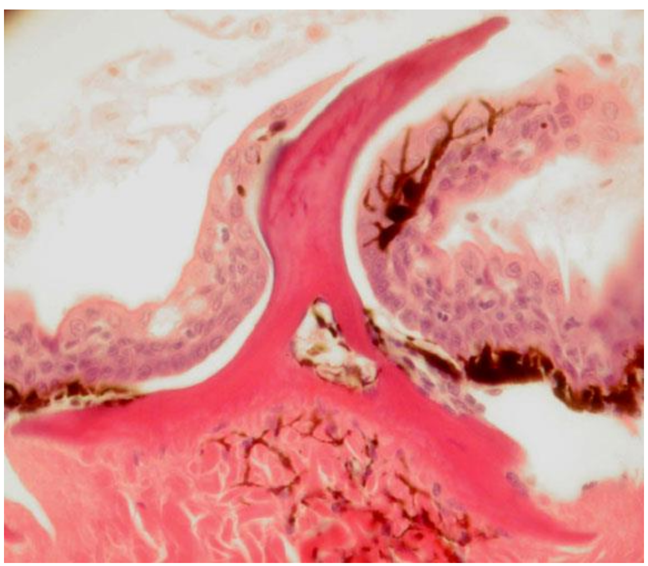
composed with dentine

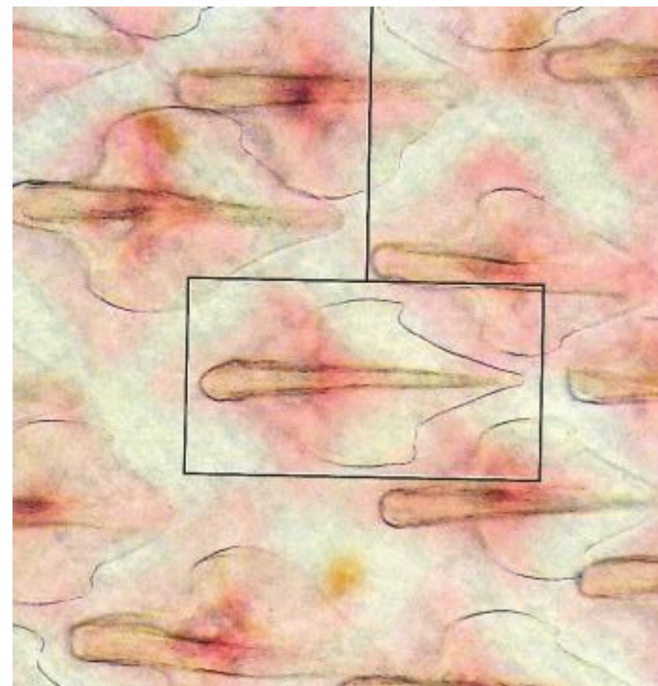
- Spine covered by enamel - contains a central pulp cavity of blood vessels, nerve endings, an lymph channels from the dermis

- Modified placoid scales forms – shark teeth, dorsal fin spines , barbs, saw teeth, some gill rakers

Placoid Scales

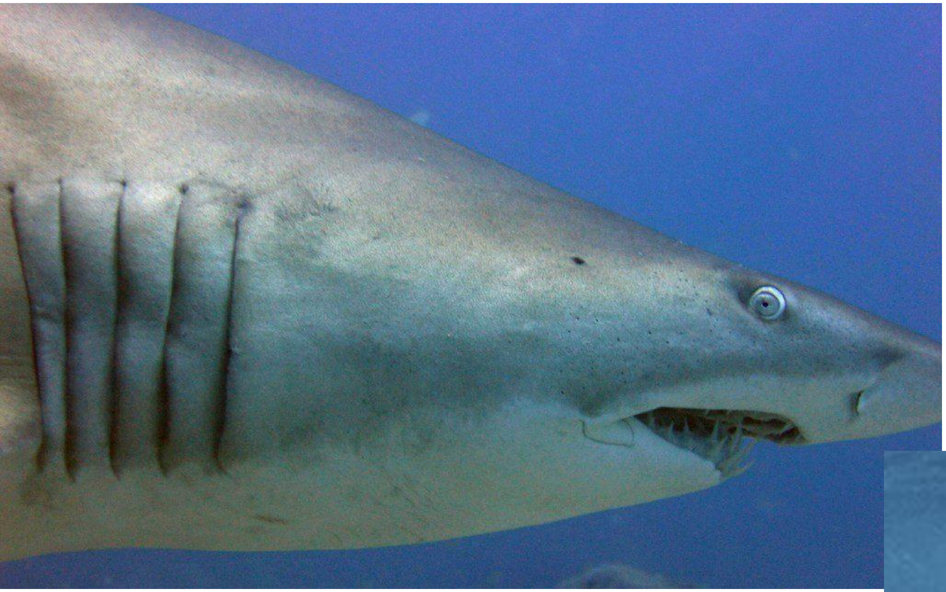






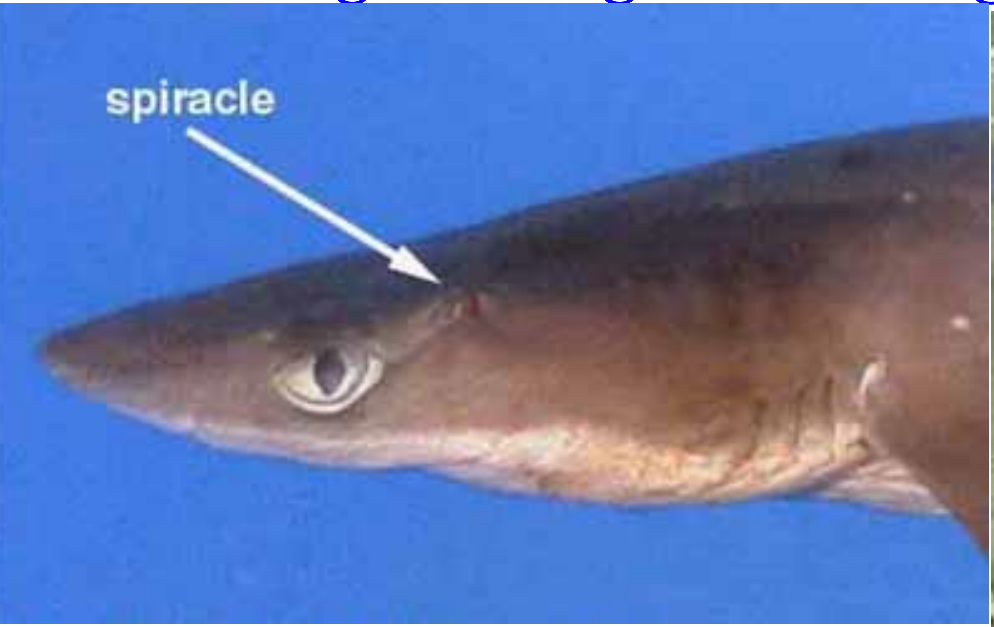
✓ Upper jaw - not fused to cranium

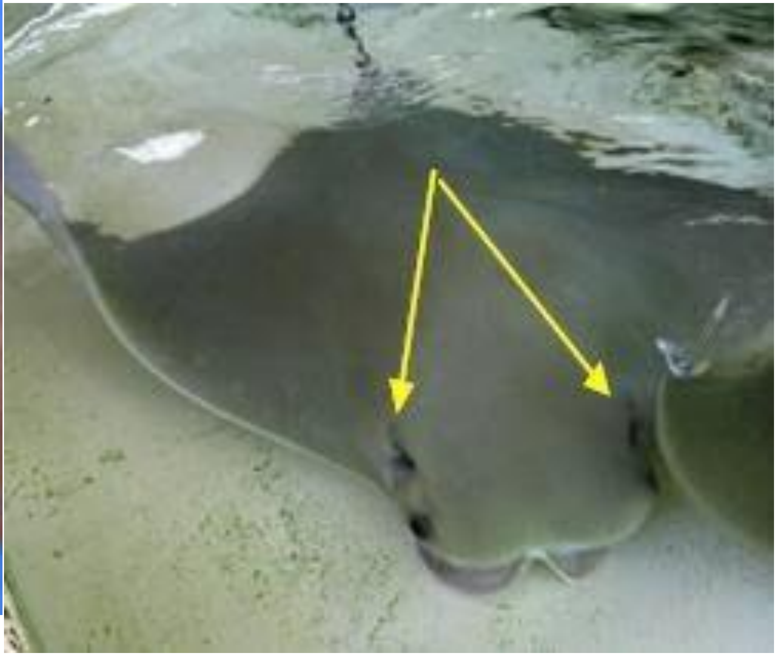
✓ Gills - Five to seven gill arches and gills in separate clefts



Spiracles on Sharks and Rays

- Spiracles - breathing openings found in some sharks and all rays & skates.
- A pair of openings located just behind the fish's eyes.
- They draw oxygenated water in from above, without having to bring it in through the gills.





- The spiracles open into the mouth and the water is passed over the gills for gas exchange and out of the body.
- The spiracles aid the fish in breathing even when it is lying on the ocean bottom or buried in the sand.
- Spiracles are absent in the requiem shark, hammerhead shark, and chimaeras.

Diversity of Elasmobranchs

Sharks

- Dominate coastal waters and pelagic sea
- Most are predators
- Few suspension feeders

Whale shark – largest fish about 12m in length

Thresher sharks - long upper tail lobe

Hammerhead shark

with flattened head with hammer like lobes

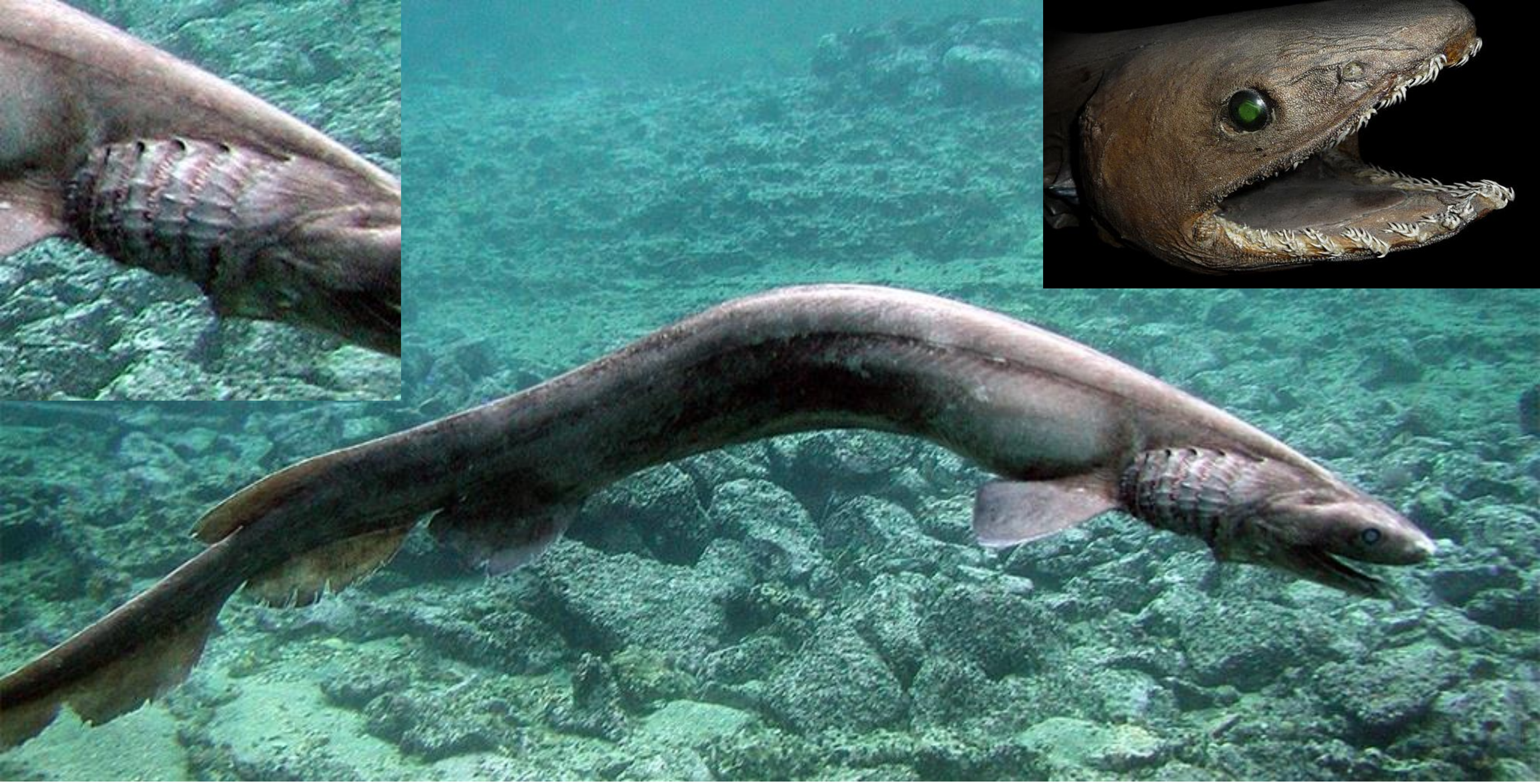
Dorso-ventrally flattened sharks

Carpet shark

Angel shark

Shark with elongated upper jaw – Saw shark

- It has a pair of barbels sticking out about halfway along the snout
- It has gill slits on the side

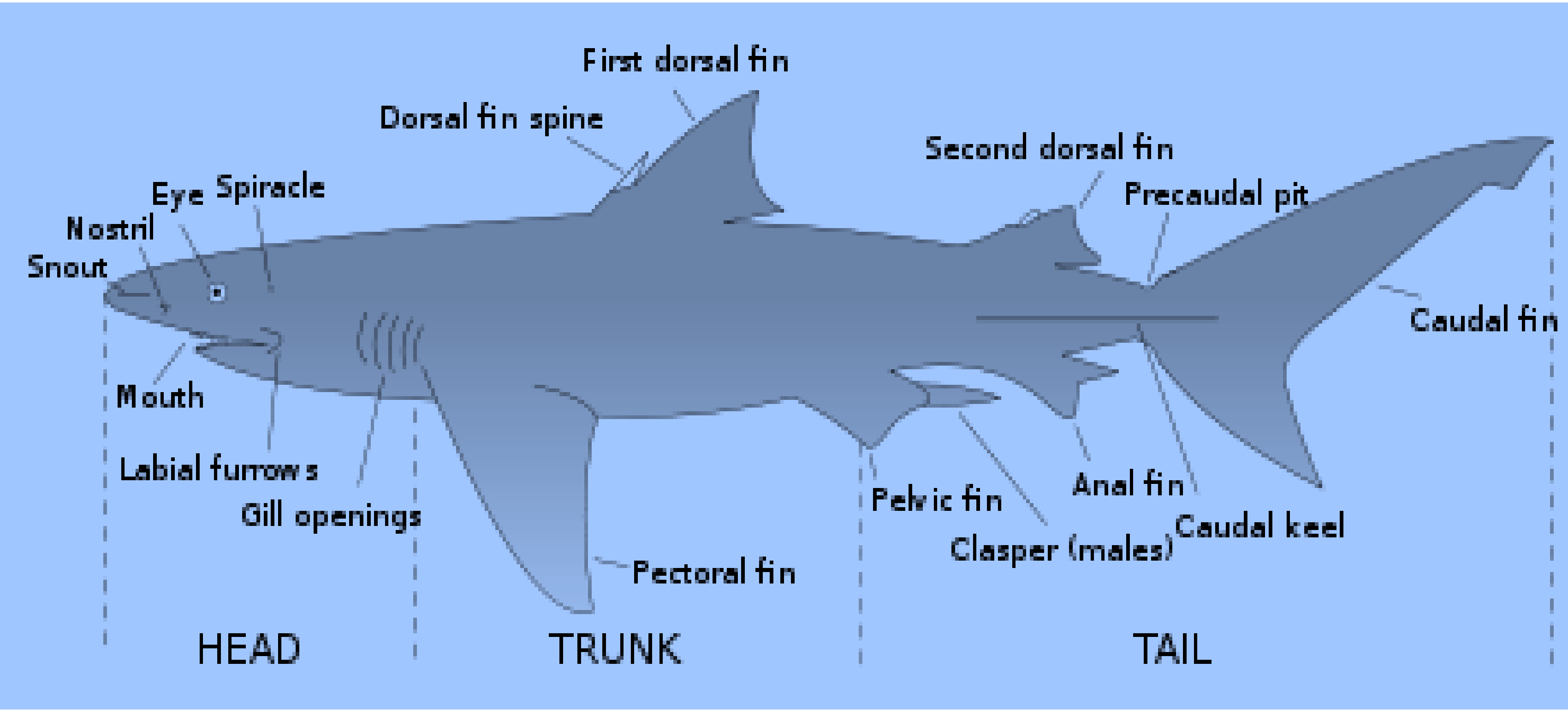


Frilled shark

- has often been termed a "living fossil".
- common name comes from the frilly or fringed appearance of its six pairs of gill slits

General morphology of shark

- Most of them have spindle shape body eg. dogfish shark; anterior pointed rostrum with ventral mouth



- Skin - Leathery skin with tooth like placoid scales
- Nostrils - paired nostrils (blind pouches) are ventral and anterior to the mouth
- Eyes - Lateral lidless eyes
- Spiracles - a pair of spiracle behind the eye
- Fins
  - ✓ paired pectoral and pelvic fins - supported by appendicular skeletons,
  - ✓ one or two median dorsal fins
  - ✓ single anal fin and a caudal fin
  - ✓ caudal fin - heterocercal tail– upper lobe is elongated than lower lobe
- Gills - five pairs of gill slits found anterior to each pectoral fin.

Predatory adaptations

- Track prey using highly sensitive senses from a kilometer or far away with their large olfactory organs.

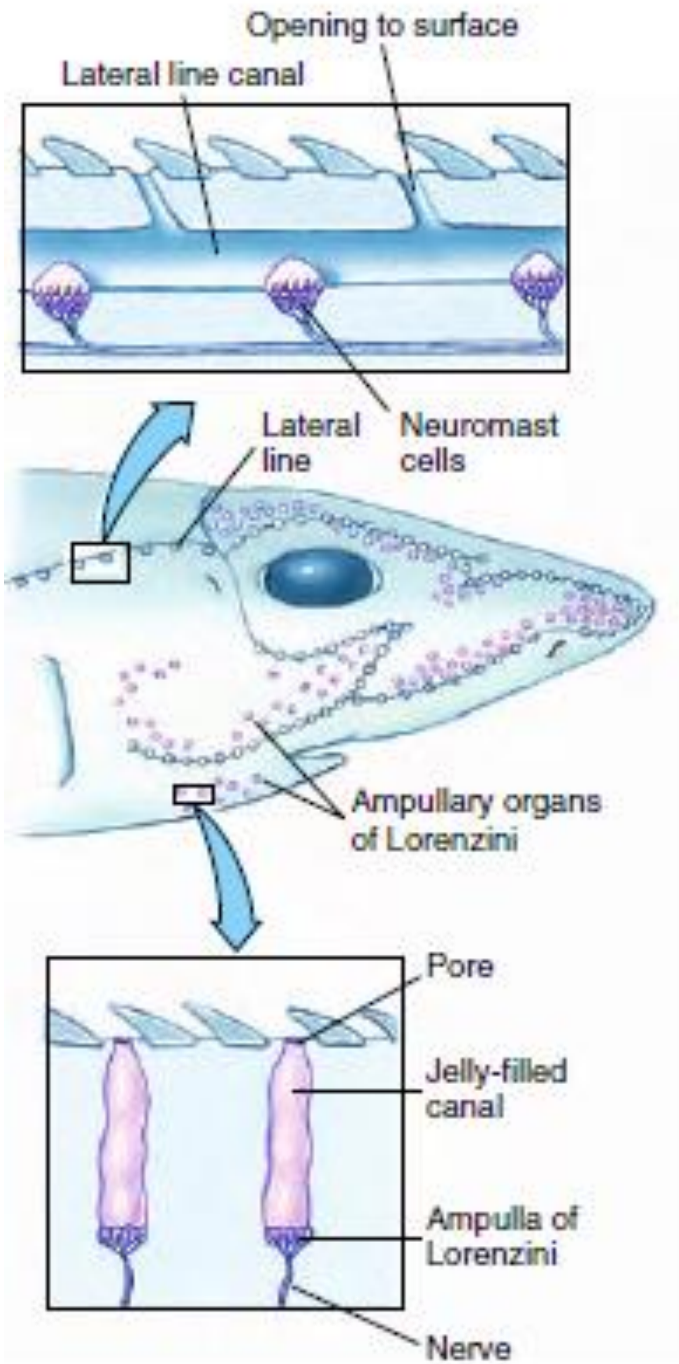
- Lateral line - mechanoreceptors present

  - sense low frequency vibrations

  - consists of *neuromasts* extend along the sides of body and over head

- Vision - excellent, even in dim light

  – can trace the prey



Sensory canals and receptors in a shark.

Ampullae of Lorenzini

- located on the head
- respond to weak electric fields : *electroreceptors*
- possibly to temperature, water pressure, and salinity.

Neuromasts

- lateral line sensors
- sensitive to disturbances in the water
- enabling the shark to detect nearby objects by reflected waves in the water.

- During the final stage of attack, sharks are guided to their prey by the bioelectric fields that surround all animals.

- In addition, sharks may use electroreception to find prey buried in the sand.

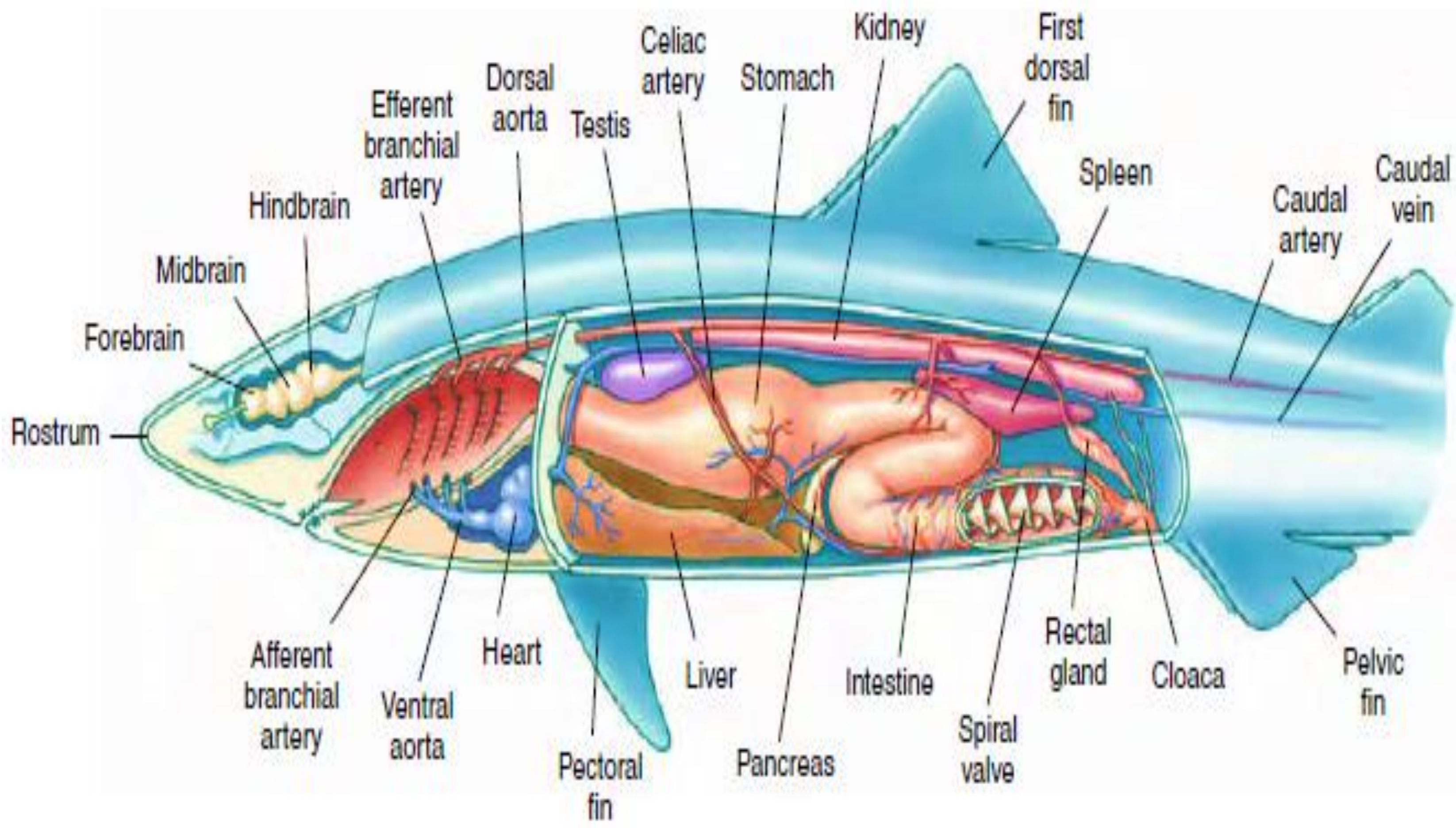
- Teeth - strong jaws with many sharp triangular teeth

  - new teeth develop inside lower jaw and move forward to replace lost teeth

Alimentary canal

- Mouth cavity - opens into the large pharynx, which contains openings to the separate gill slits and spiracles.
- Esophagus – wide, , runs to the stomach
- Stomach - J shaped , short
- Intestine - short, straight, contains the spiral valve that slows passage of food and increases the absorptive surface.
- Liver and pancreas - open into the intestine
- Rectum - short rectum

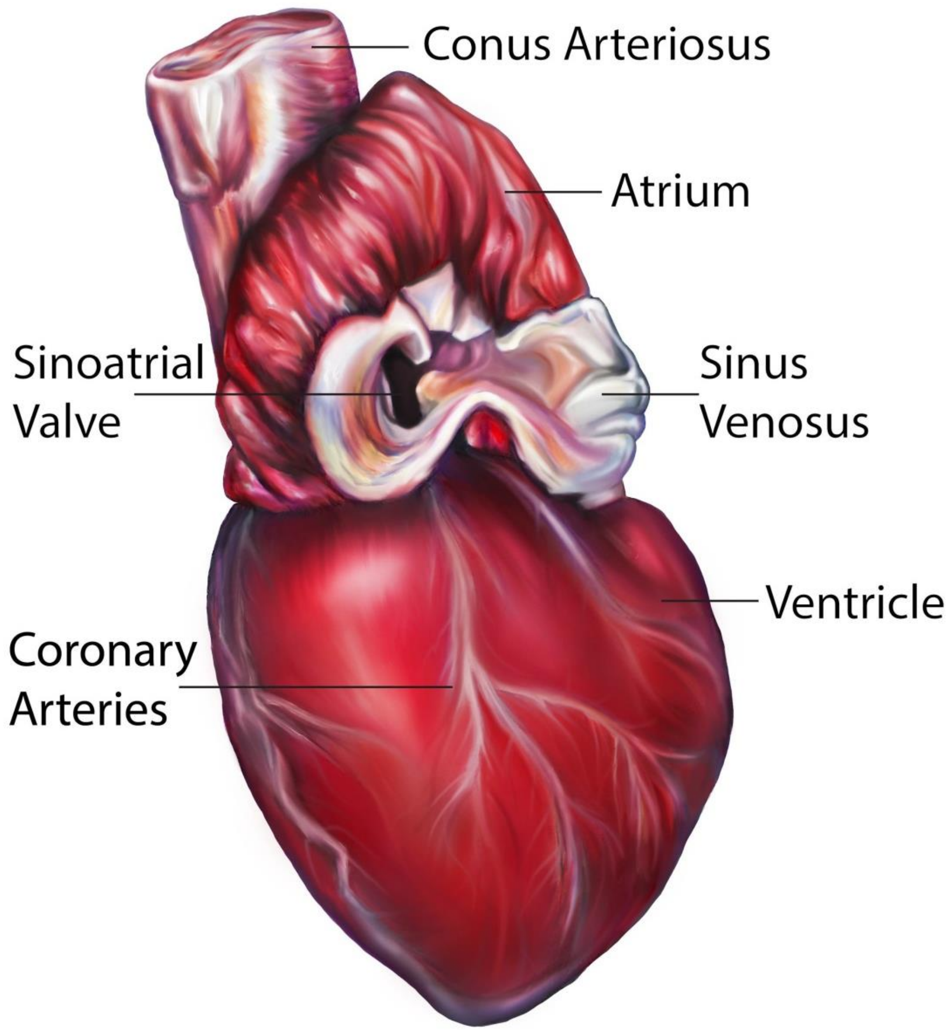
- Rectal gland - unique to chondrichthyans, which secretes a colorless fluid containing a high concentration of sodium chloride.
- Opisthonephric kidney - rectal gland assist in regulating the salt concentration of the blood



Heart

- The chambers of the **heart** are arranged in tandem formation – 1 atrium and 1 ventricle

- The flow pattern of the circulatory system is basically the same as that of other gill breathing vertebrates .



Reproduction

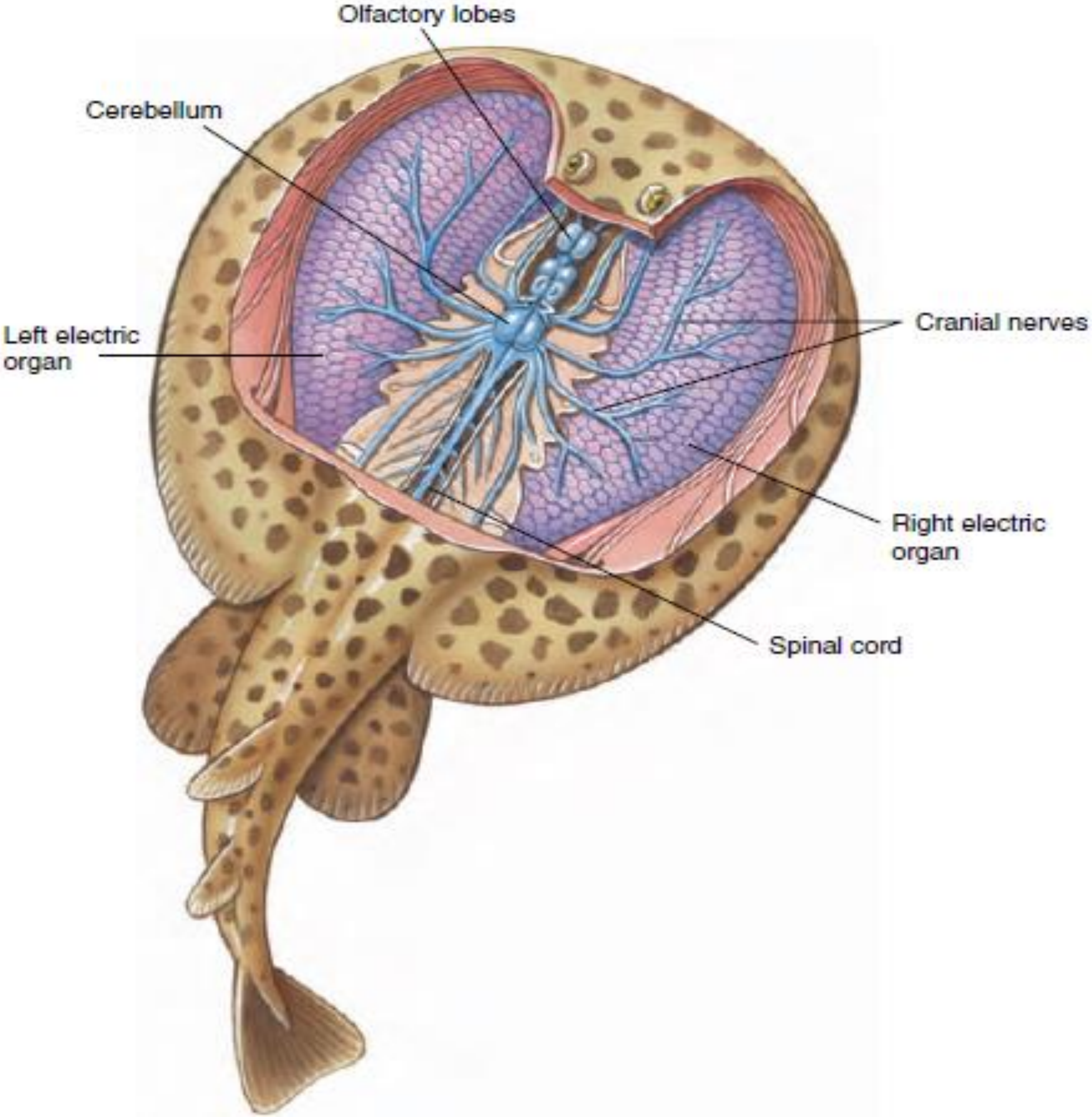
- All chondrichthyans have internal fertilization, but maternal support of the embryo is highly variable.

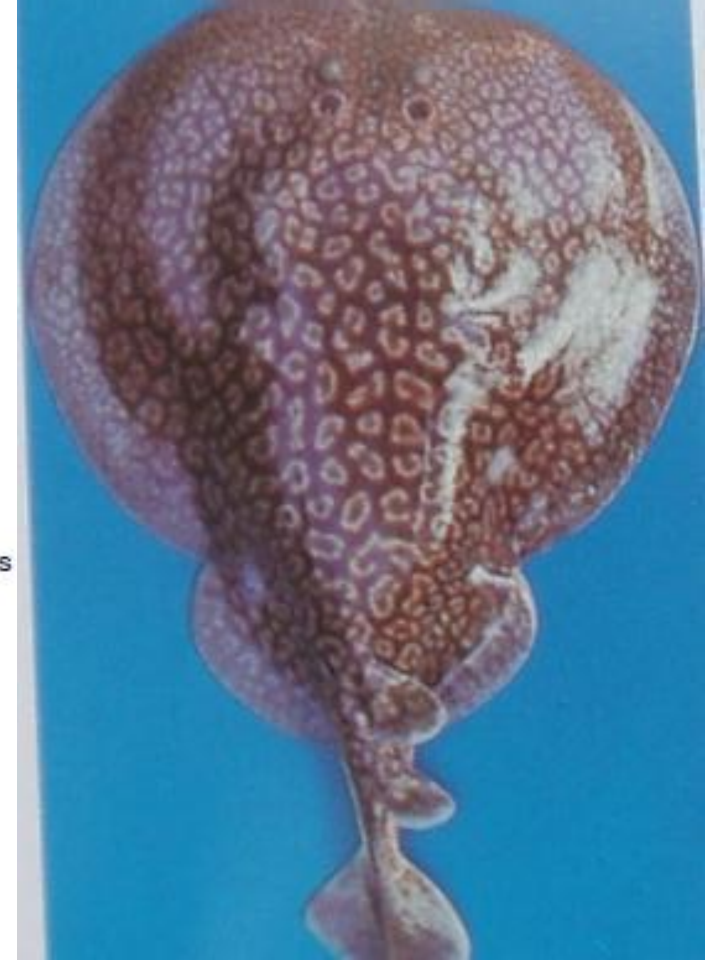
- They can be
  - ✓ oviparous - lay large, yolky eggs immediately after fertilization
  - ✓ ovoviviparous - developing young in the uterus while they are nourished by the contents of their yolk sac until born.
  - ✓ viviparous – retain the developing young in the uterus while they are nourished by the contents of their yolk sac until born.

- Nourishment for growing embryo
  - ✓ embryos receive nourishment from the maternal bloodstream through the placenta, or from nutritive secretions, “uterine milk,” produced by the mother.
  - ✓ Some sharks (sand tigers) exhibit a grisly type of reproduction in which embryos receive additional nutrition by eating eggs and siblings.
  - ✓ However, regardless of the form of maternal support, once the eggs are laid, or the young born, all parental care ends.

Skates and Rays

- Habitat :Distributed mainly in marine but some rays are found in freshwater.
- Specialized for bottom dwelling
- Sting rays – Slender whip like tail with one or more saw edged spines and venom glands at base
- Electric rays- large electric organs on each side of head

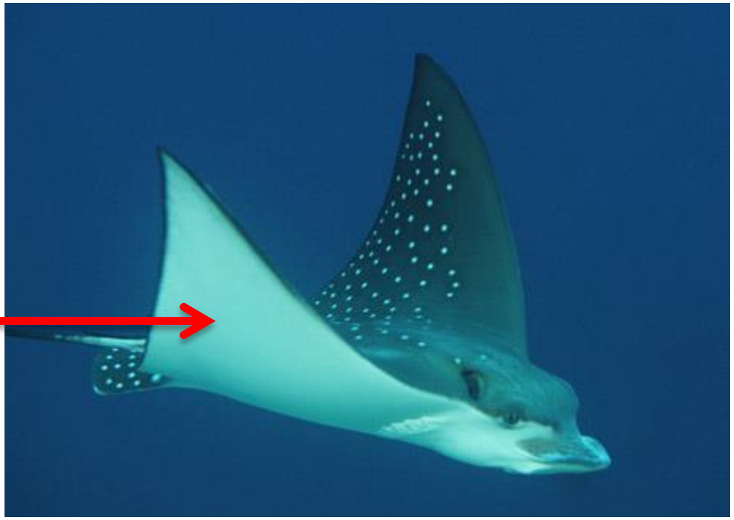

Olfactory lobes
Cerebellum
Left electric organ
Cranial nerves
Right electric organ
Spinal cord



General morphology of skates and rays

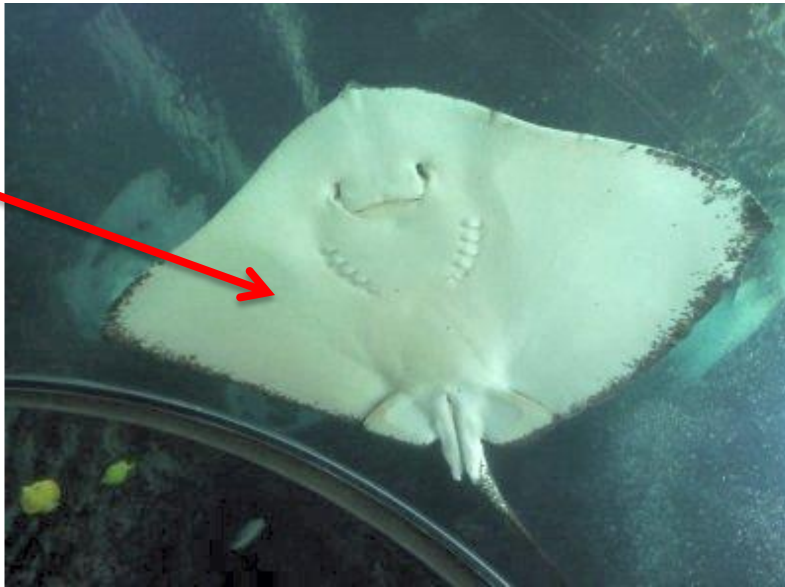
- Dorsoventrally flattened body

- Enlarged pectoral fins that are fused to head



- A pair of dorsal spiracles present on the top of the head which takes water in and transferred to the gills

- Mouth and five pairs of gills present ventrally



Ray

unilobed
pelvic fin

long whip like
tail with spines
& without fins

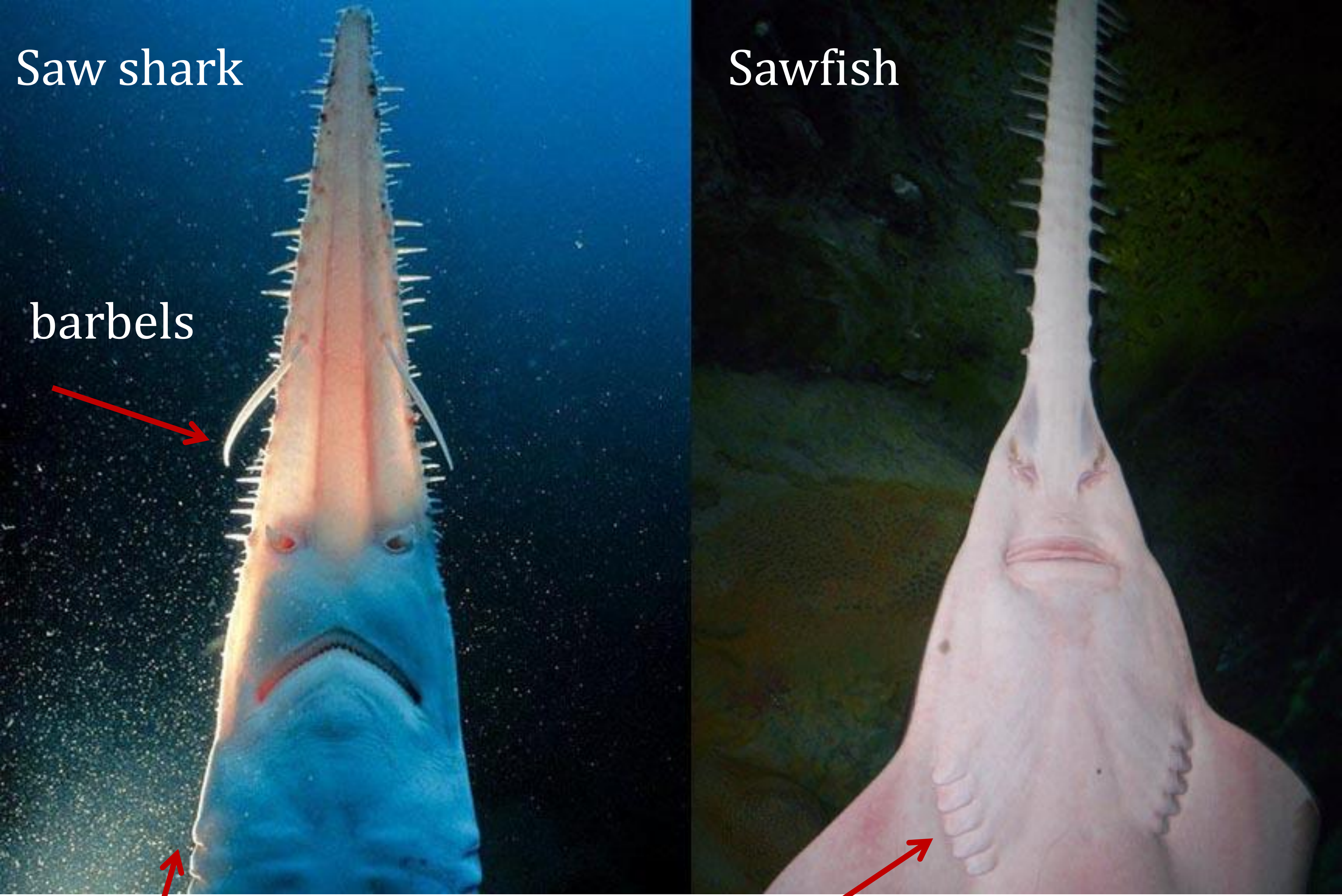
Skate

bilobed
pelvic fin

short tail with
small caudal &
dorsal fins

Sawfishes

- also known as carpenter shark – is a family of ray
- characterized by a long, narrow, flattened rostrum or nose extension, lined with sharp transverse teeth, arranged in a way that resembles a saw.
- Several species of sawfishes can grow to about 7 m .
- All species of sawfishes are listed as Endangered or Critically Endangered by the IUCN
- They face the threat of extinction as a result of habitat loss and overfishing.



Lateral gill slits

Ventral gill slits

Subclass - Holocephali

- diverge from the earliest shark lineage 360 million years ago
- confined to deep sea

Characteristics

- Mouth - jaws bear large grinding flat plates instead of toothed mouth
- Jaw - upper jaw completely fused to the cranium
- Skin - smooth and no scales
- Colour - range from black to brownish gray

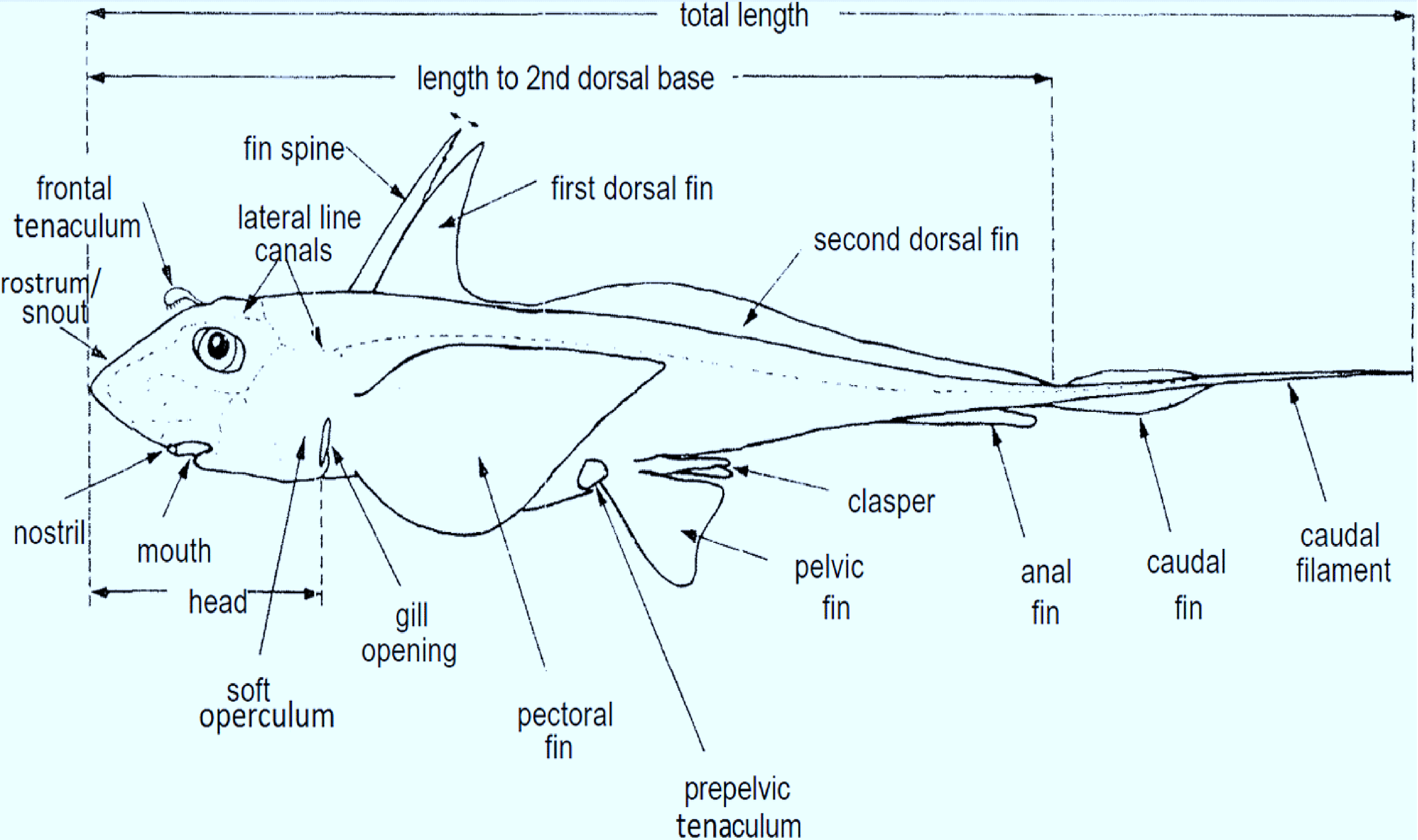


diagram of a male chimaera

(FAO.org)

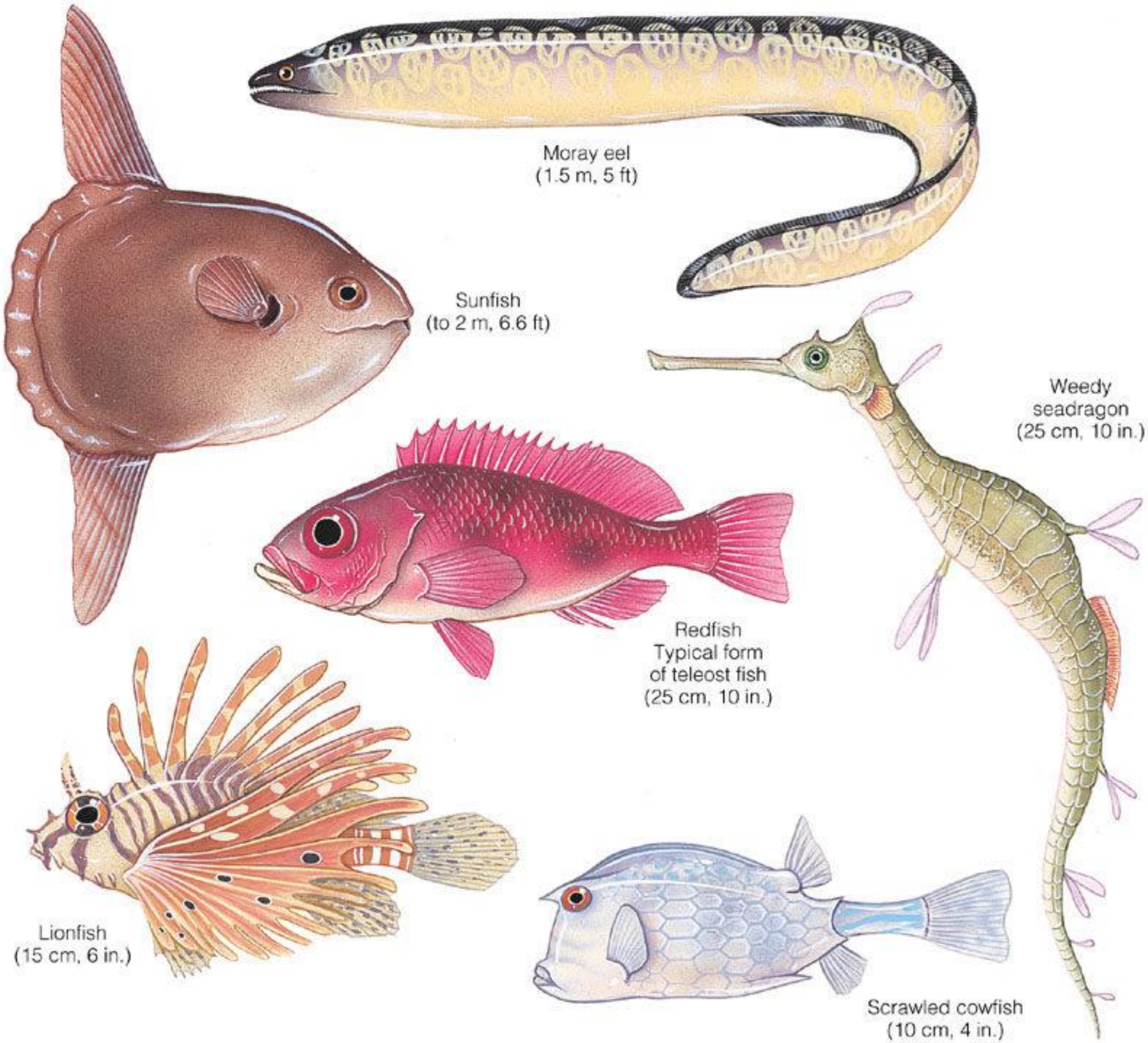
- Gills - four gill slits covered by operculum – single opening
- Fins - two dorsal fin , the first of which in many species may be armed with a retractable venomous spine
- Clasping organ - accessory clasping organs
  - tentaculum in males

e.g. *Chimaera* (ghost shark/ratfish)

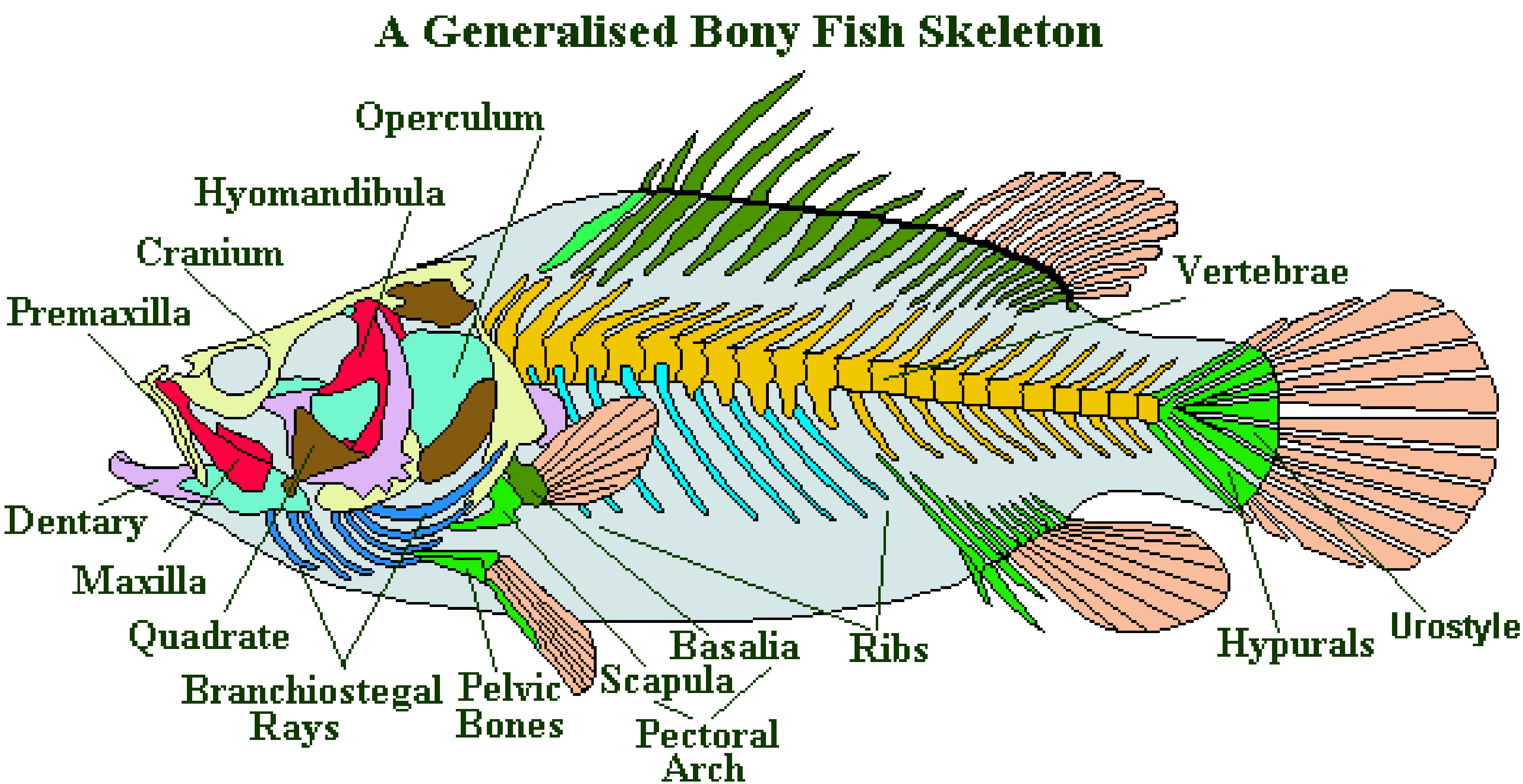

Bony Fish : Osteichthyes


General characters of bony fish

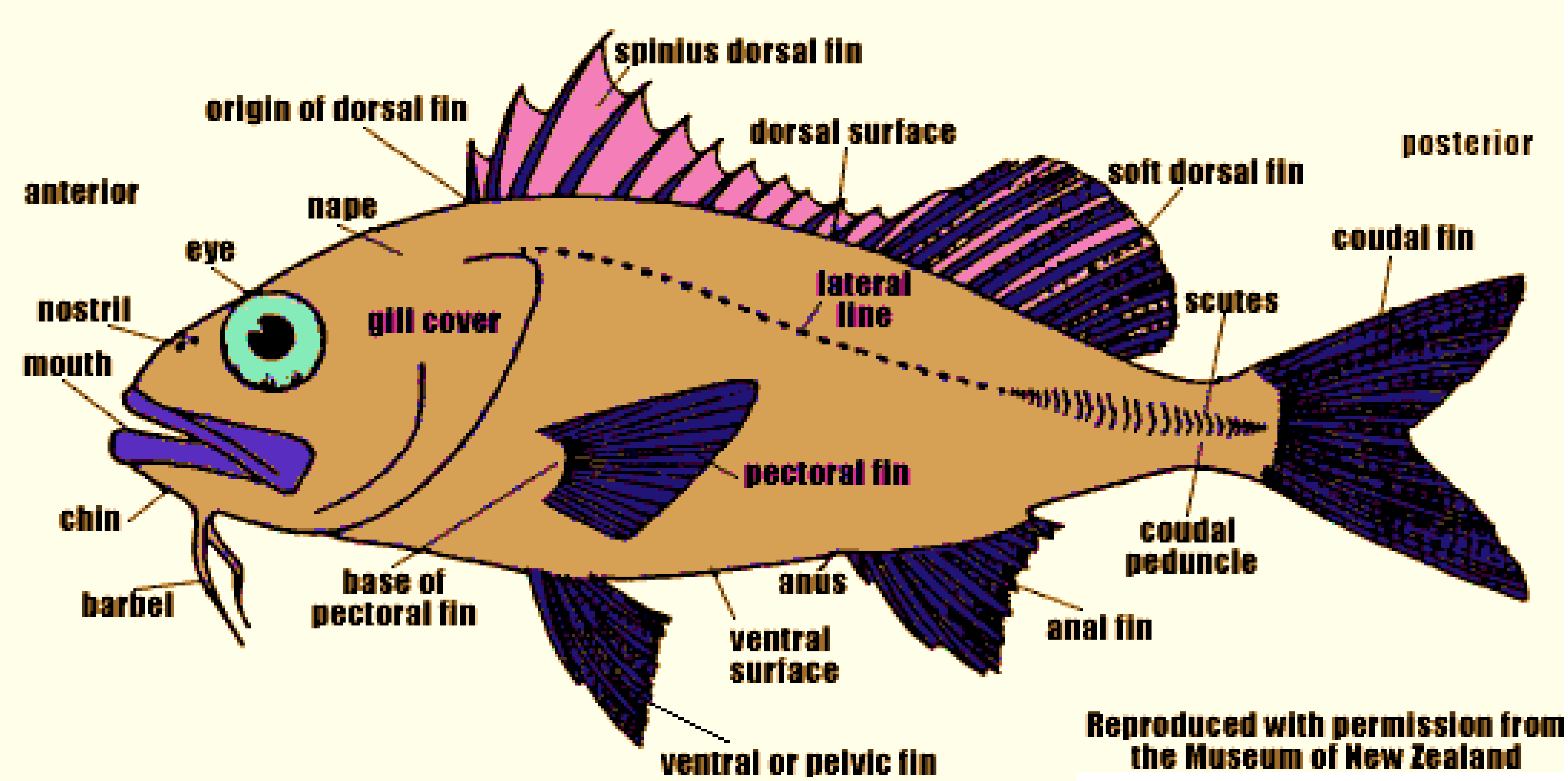
- Body shape - usually spindle shaped – but diverse



- Endoskeleton - partly or wholly bony



- Fins - Median and paired fins
- Gills - gill slits are covered with operculum
- Nostrils - located on the dorsal side of snout



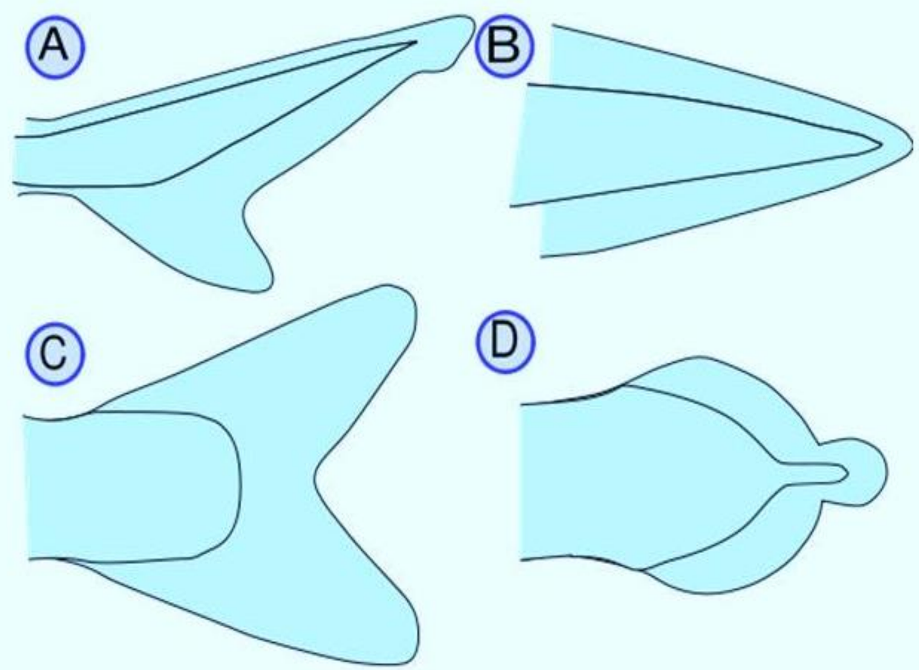
A - Heterocercal - vertebrae extend into the upper lobe of the tail, making it longer eg. sharks

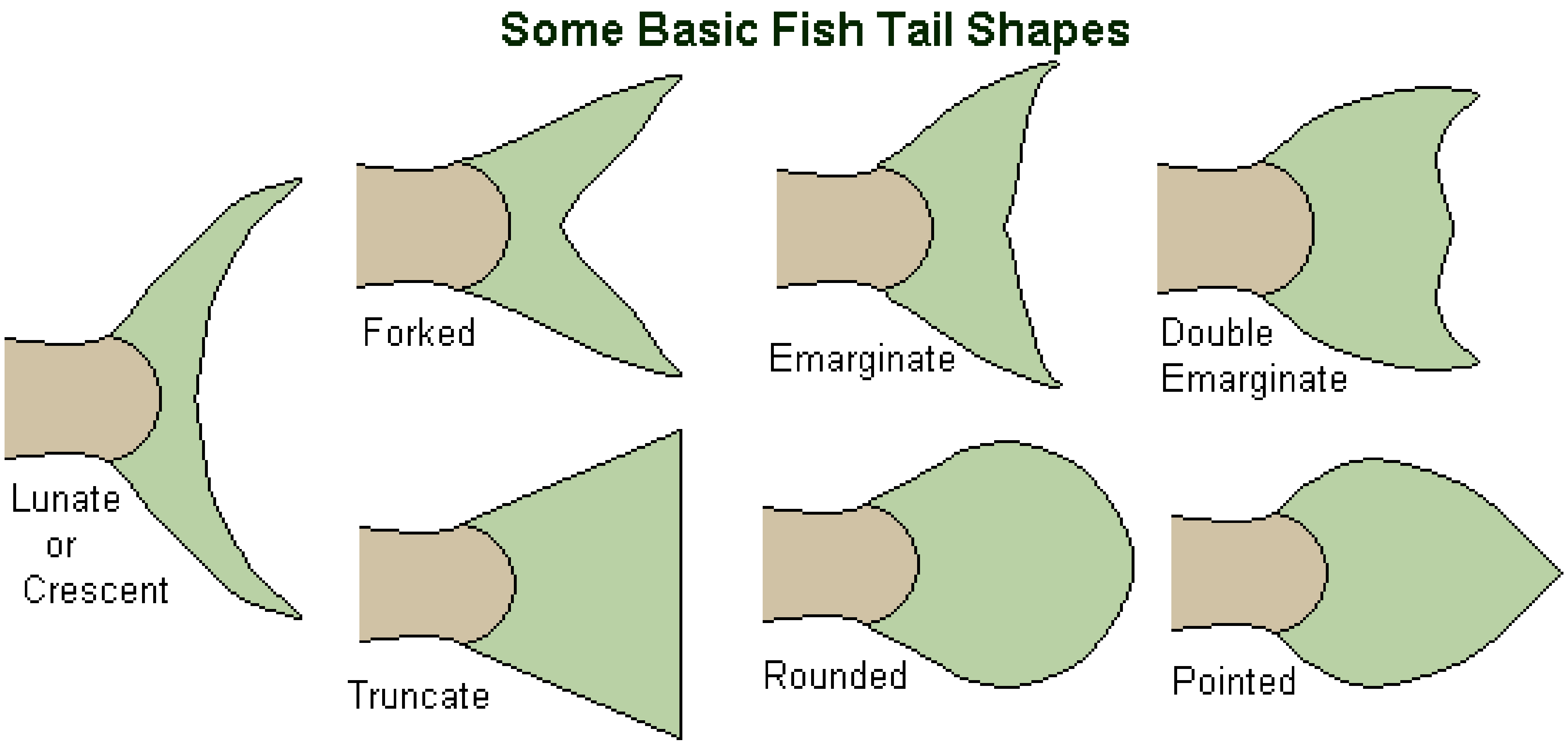
B - Protocercal -the vertebrae extend to the tip of the tail and the tail is symmetrical but not expanded eg. amphioxus

C - Homocercal - fin appears superficially symmetric but the vertebrae extend for a very short distance eg. tuna

D - Diphycercal - the vertebrae extend to the tip of the tail and the tail is symmetrical and expanded eg.bichir, lungfish, lamprey ,coelacanth

Different types of caudal fins classified according to its extension of vertebral column





Homocercal fin

Scales

- The scales of bony fishes evolved a long time ago
- The size, and distribution of scales over a fish's body often, but not always, reflect the way it lives.
- Fish that swim quickly, or that live in fast flowing waters have small scales eg. Trout, Tuna
- Fish that swim slowly or in slow moving waters tend to have larger scales eg. Carp.

Tuna

Common carp

- In some fish the scales have become modified into bony plates to make armour eg. S. American Catfishes ,Sea horses and Pipe fish.
- In these cases the added protection is paid for by reduced flexibility and speed.
- The scales of some fish decrease in size from the head towards the tail provides greater flexibility towards the tail of the fish eg. Carp and Herring
- Scales are modified to moveable spines eg. Porcupine-fishes (Diodontidae) , Puffers (Tetraodontidae)

Seahorse

Pipefish

Porcupine fish

Puffer fish

- Different types of scales are seen in bony fish
  1. Cosmoid scales
  2. Ganoid scales
  3. Elasmoid scales(Ctenoid and Cycloid scales)

1. Cosmoid scales

- Ancient scales had four layers,
  - dense bone,
  - spongy bone,
  - dentine
  - enamel
- Cosmoid scales only exist in the modern world on the Ceolocanth (*Latimeria chalumnae*) or as fossils.

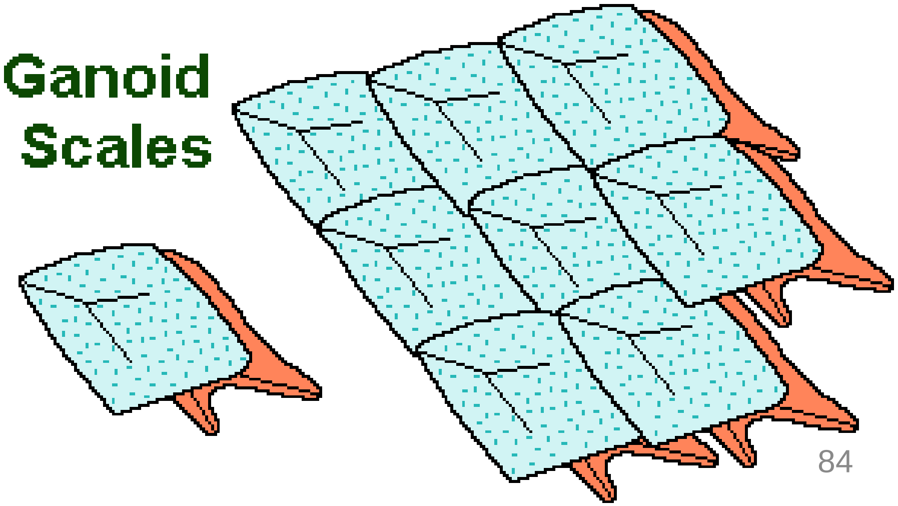
Cosmoid scale

Coelocanth

The scales of the remaining bony fishes ie. Ganoid and Elasmoid have only two layers, a calcified one and a fibrous one.

2. Ganoid scales

- Ganoid scales are derived from Cosmoid scales
- Hard solid scales
- Theses are evolutionary older style and are found on Bichirs, Gar-fish, Sturgeons and Reedfish.



Gar pike

3. Elasmoid scales

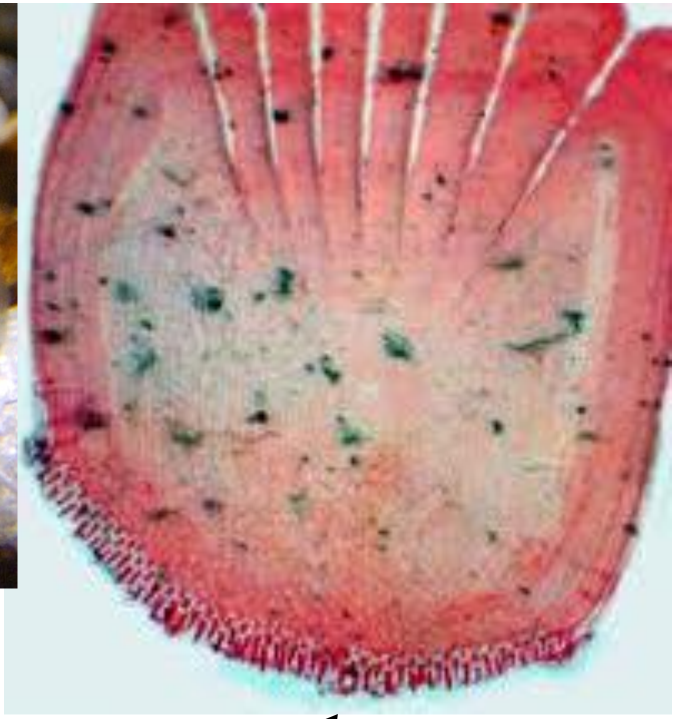
- The most common form of scale.
- It is the thin plate that you find on most fishes.
- It is often in two forms,

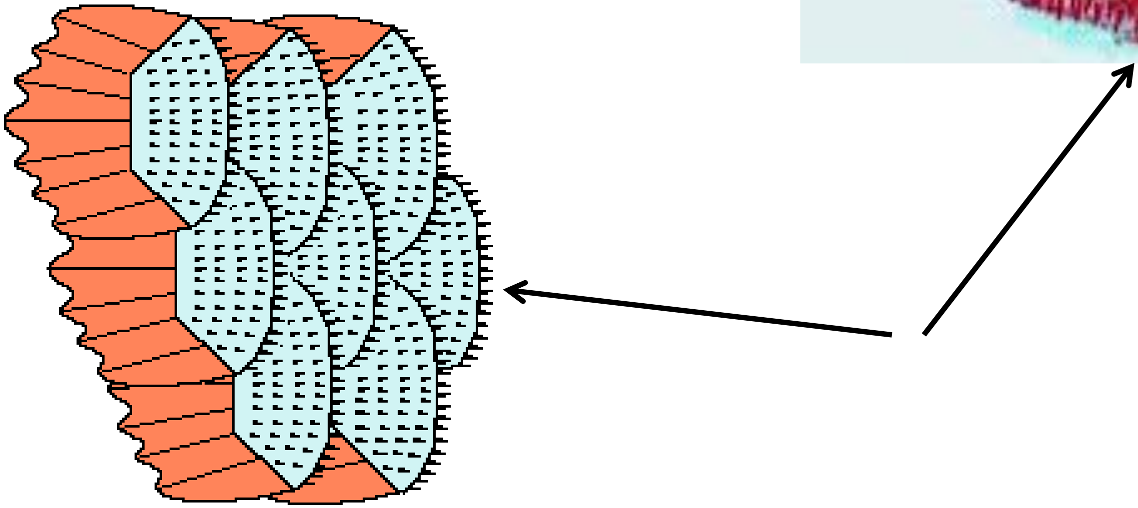
  a) Ctenoid scales

  b) Cycloid scales

a) Ctenoid scales

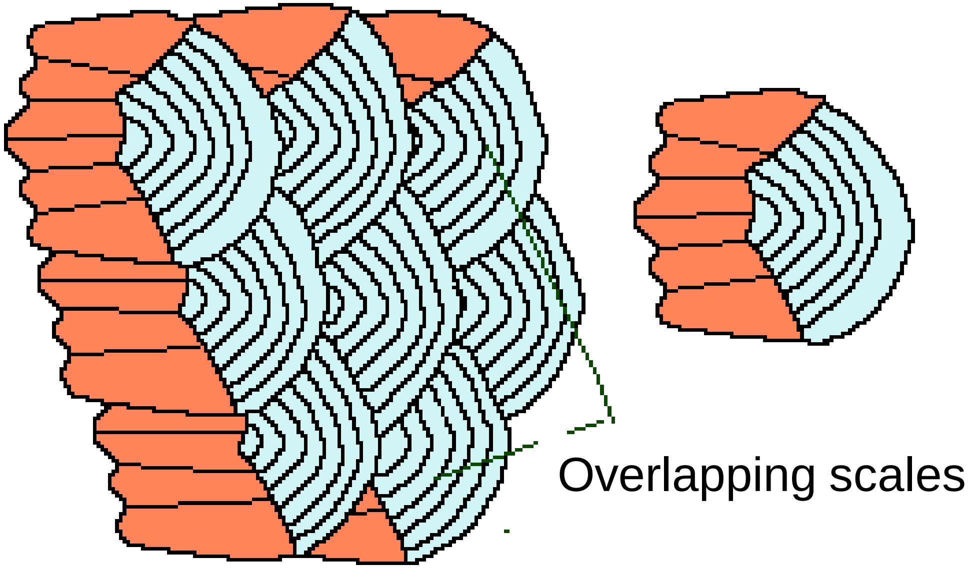
- Ctenoid scales have a set of fine teeth along the posterior edge eg. perch



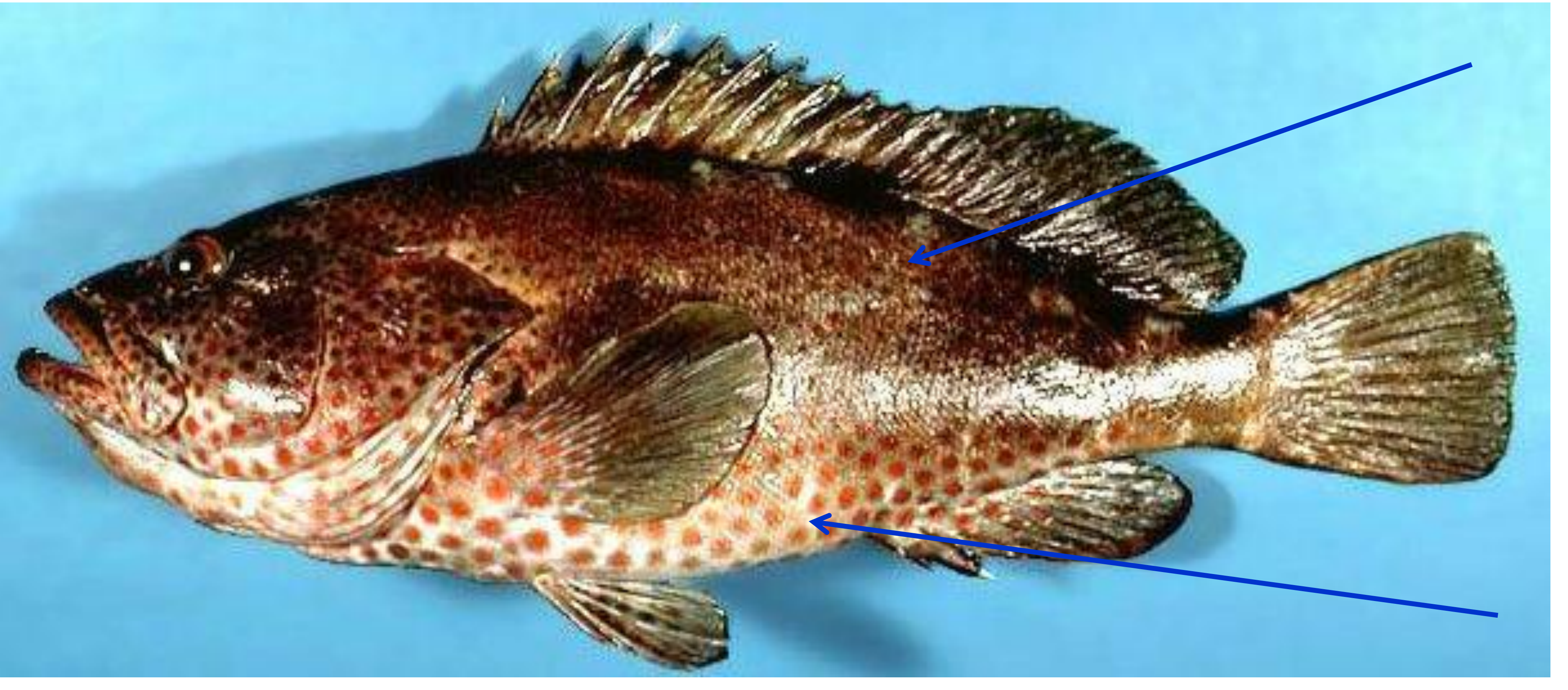


b) Cycloid scales

- Cycloid scales are simply rounded on the outer/posterior edge Eg. herrings, carp



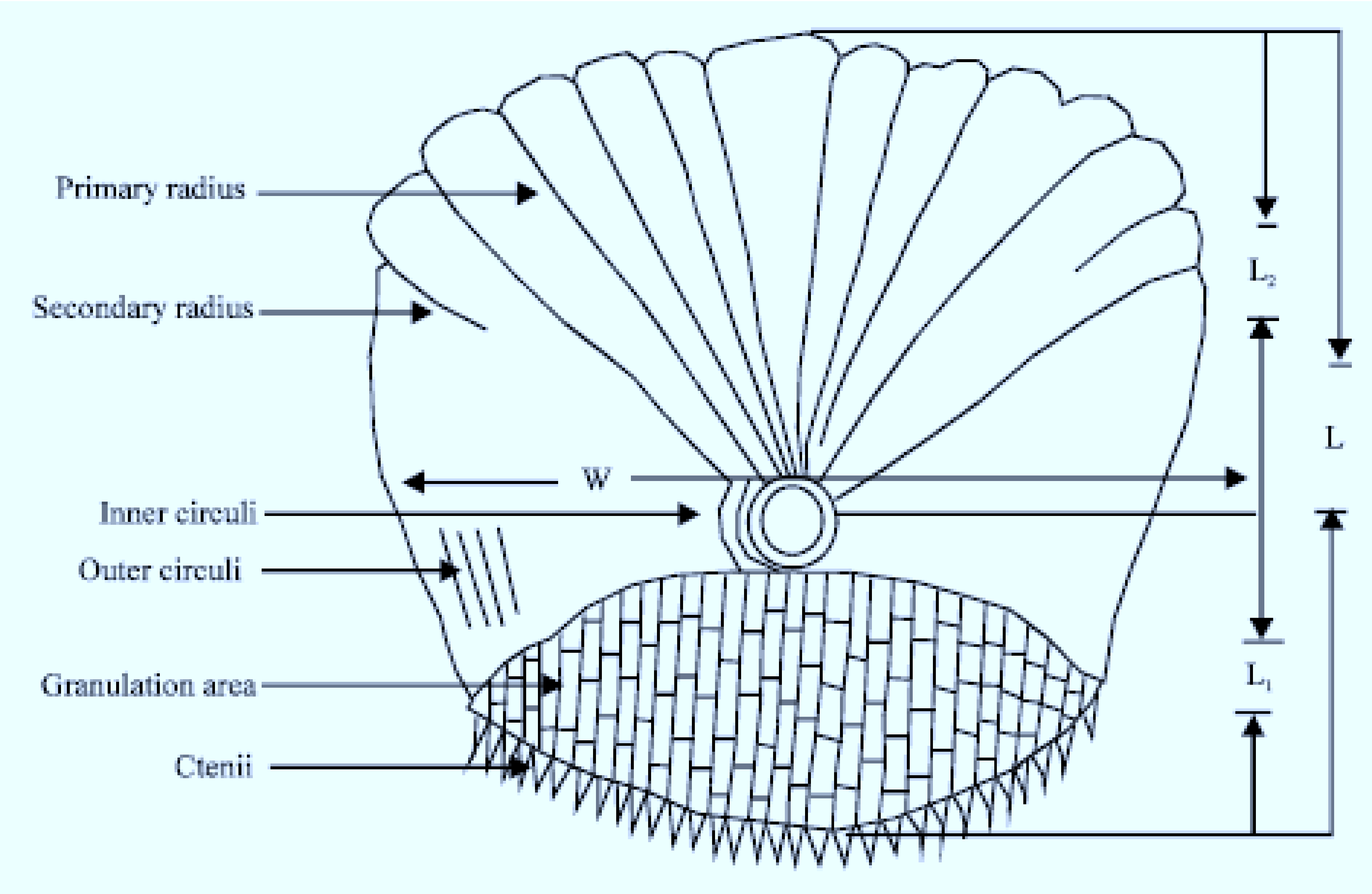
- Ctenoid scales and cycloid scales may be found on the same fish eg. Sea Perches, (*Epinephelus sp.*)
- They have mostly ctenoid scales above the lateral line and cycloid below



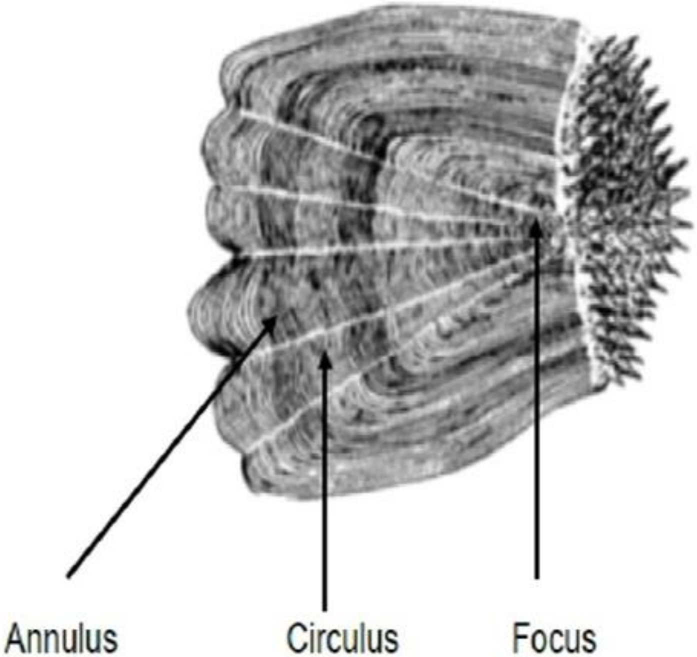
- Some species do not have any scales on the body – naked eg. Sunfish (*Mola mola*) and the Siluroidei (Naked Catfish)

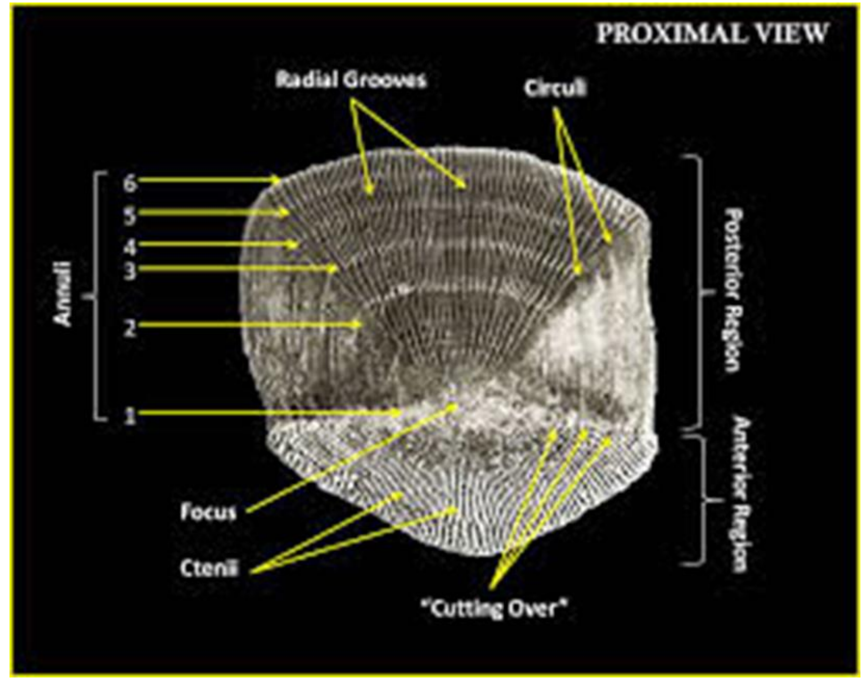
- Some fish appear to have no scales but they really have microscopic scales deeply embedded in their dermis. Eg. common eel

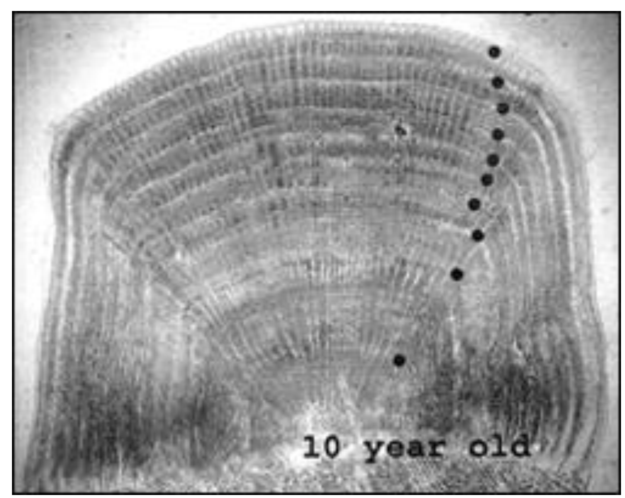
Structure of scale

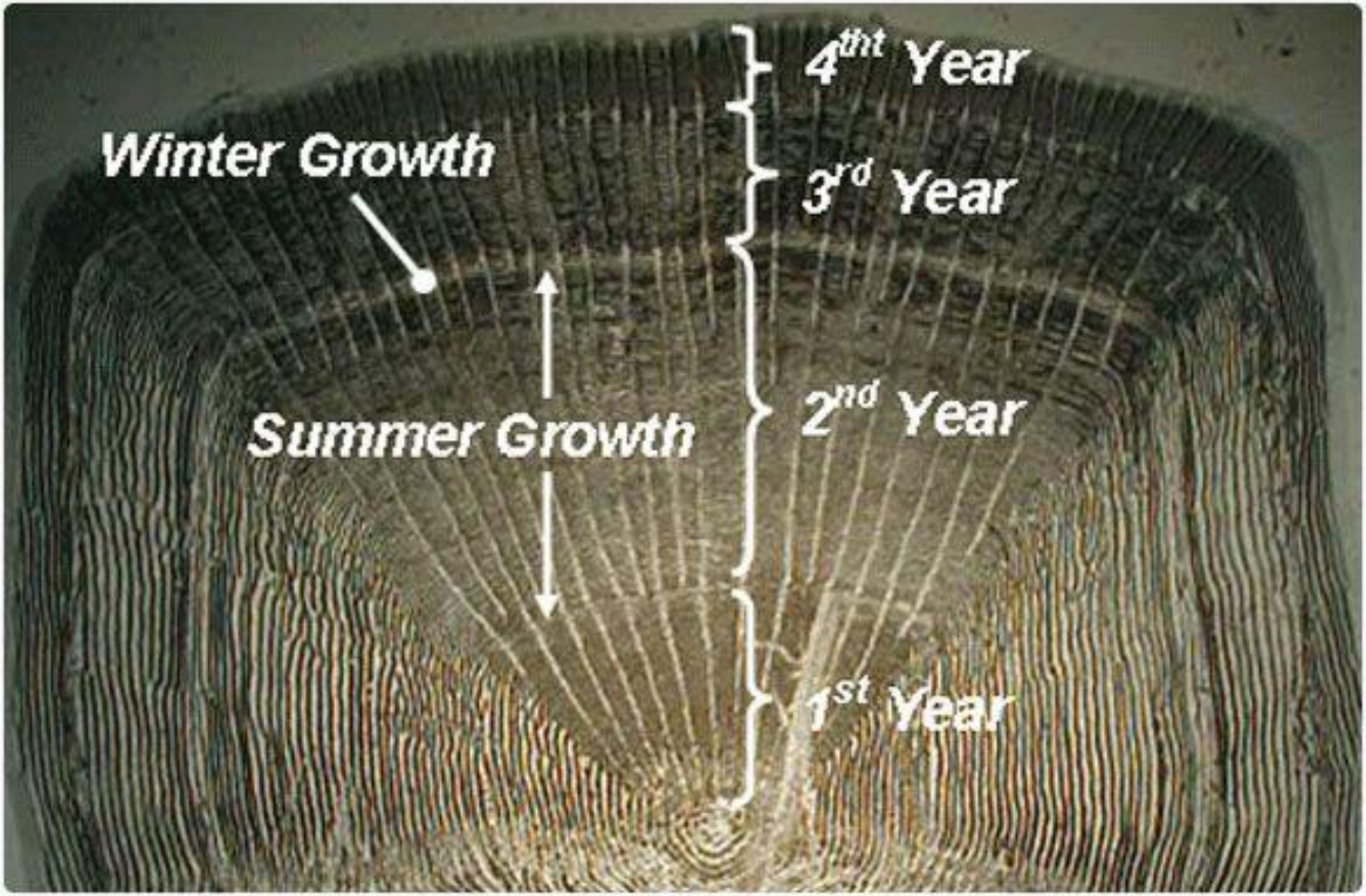


Scales are used in age determination

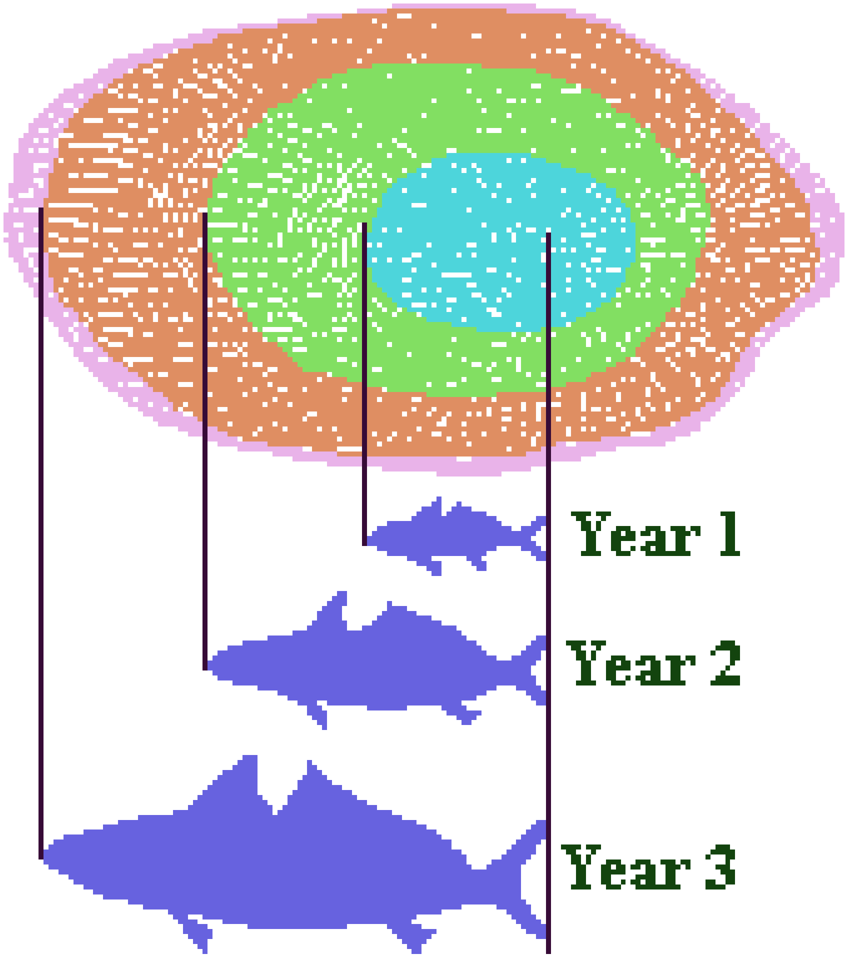







4tht Year
Winter Growth
3rd Year
2nd Year
Summer Growth
1st Year

A Diagramitic Representation of the relationship between the annual growth of a Fish's body and of its scales.



- Devonion period bony fishes radiated into two linages

1) Class Sarcopterygii (Lobe - finned fishes)

2) Class Actinopterygii (Ray – finned fishes)

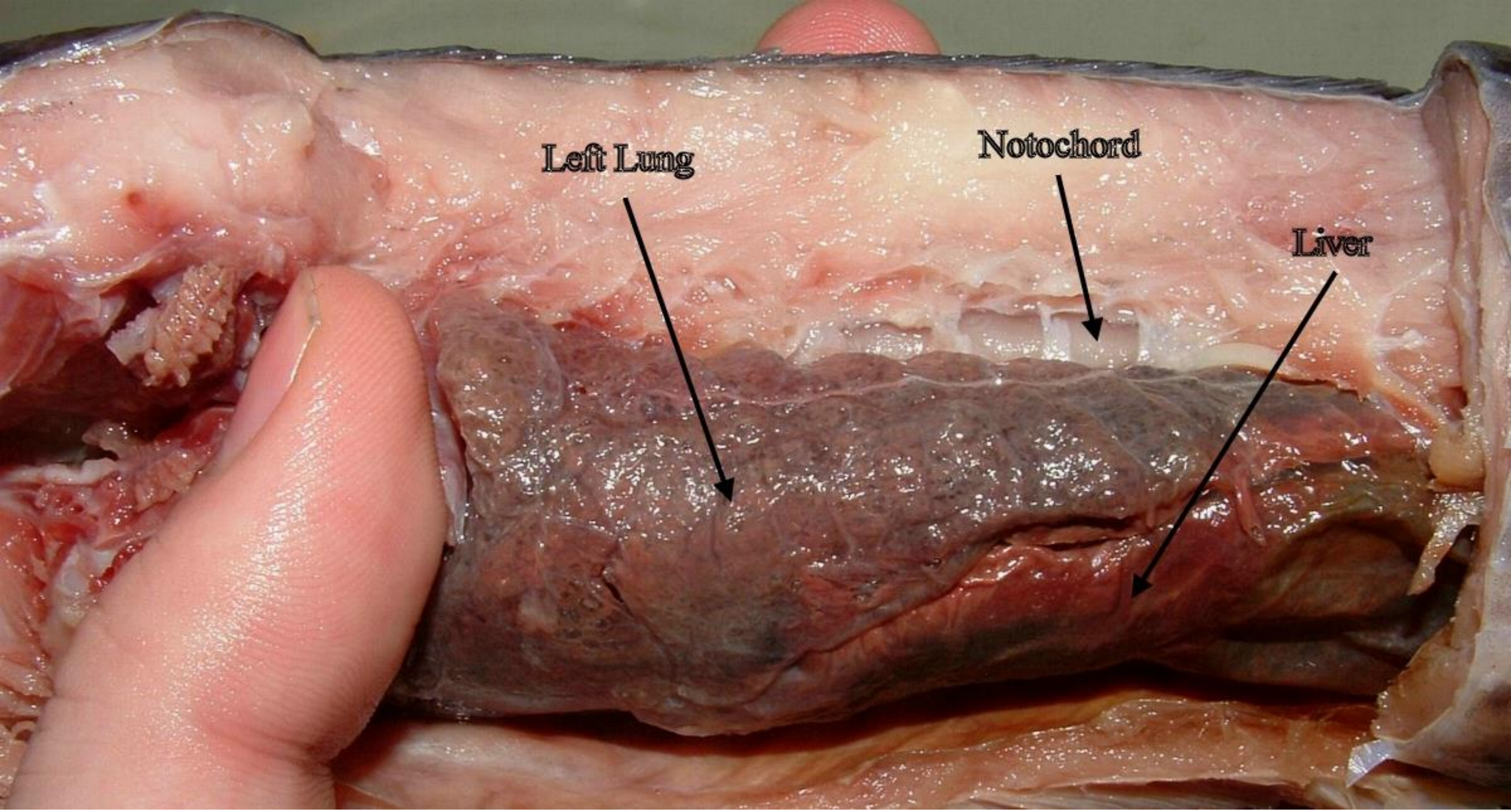
Class - Sarcopterygii

- Represented by only seven species
- Six species of lungfishes and the coelacanth
- Strong, fleshy pectoral fins used like legs to scuttle along the bottom
- Early sarcopterygians had lungs, gills and a heterocercal fin

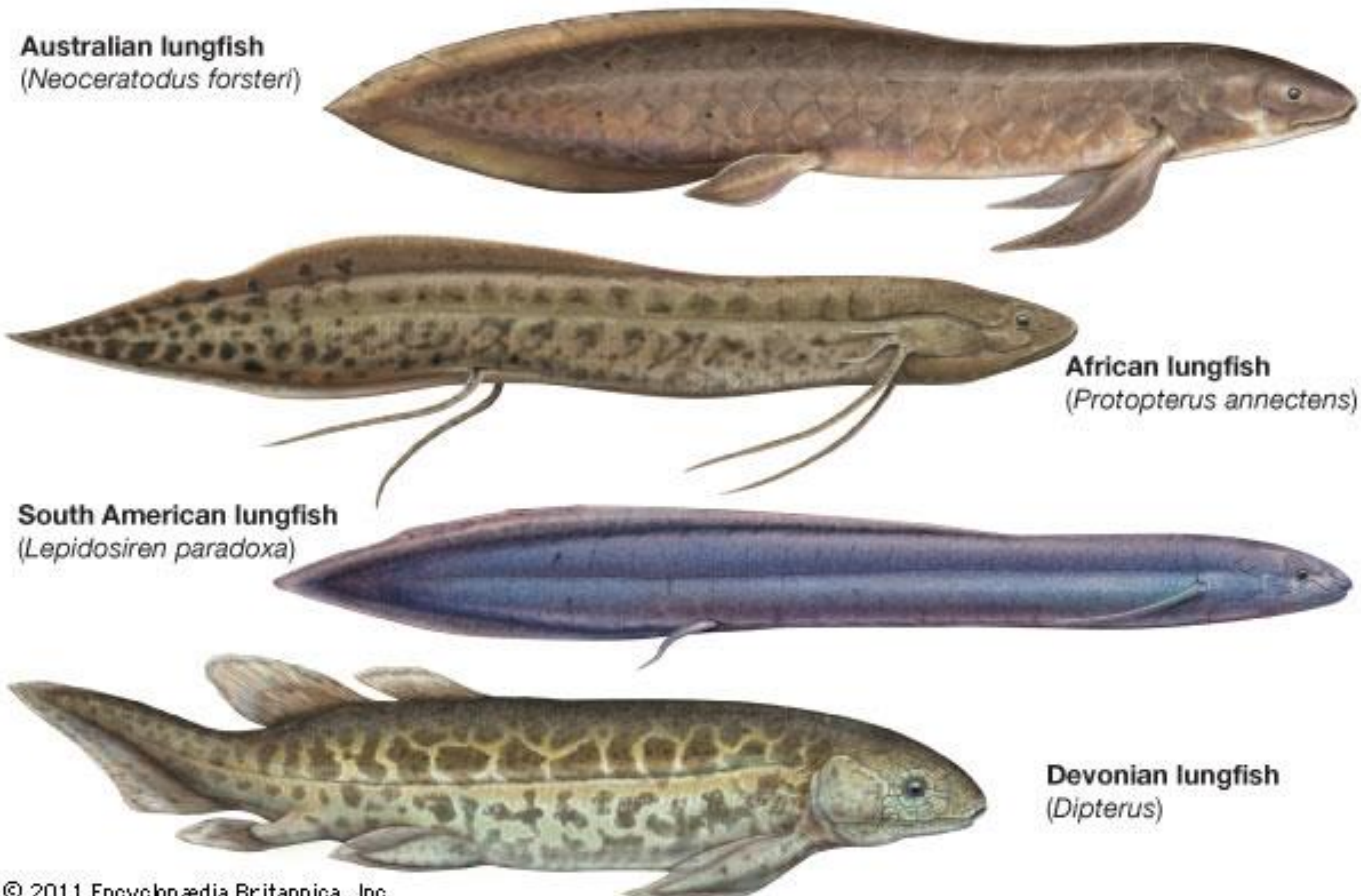
Characteristics

- Ossified skeleton
- Skin - covered with dermal scales, cycloid scales
- Fins - Paired fins with sturdy internal skeleton and musculature within limb
  - Diphycercal caudal fin in living species
- Teeth - with enamel covering (cosmin)
  - typically have crushing plates restricted to palate

- Gills - Single gill opening covered by an operculum
- Intestine - with spiral valve
- Swim bladder - Usually present a lung like swim bladder for respiration and buoyancy
- Heart - with a sinous venosus, two atria and partly divided ventricle with five aortic arches
- Circulation - double circulation with pulmonary and systemic circuits
- Nervous system - with olfactory and optic lobes
  - a cerebrum and a cerebellum
  - ten pairs of cranial nerves
- Reproduction - Sexes separate
  - fertilization external or internal



Dissection of *Protopterus dolloi*
a sarcopteryigian fish with lungs


Australian lungfish
(Neoceratodus forsteri)
African lungfish
(Protopterus annectens)
South American lungfish
(Lepidosiren paradoxa)
Devonian lungfish
(Dipterus)

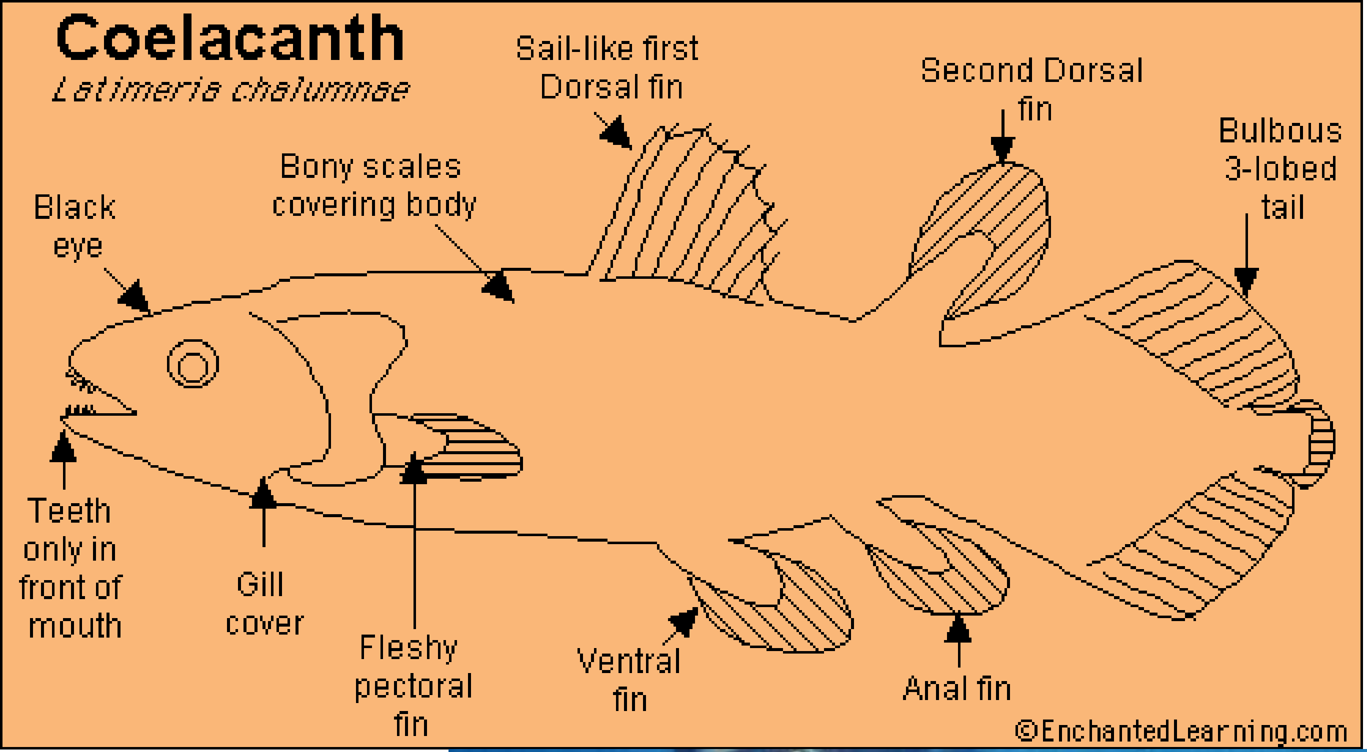
***Lepidosiren* (S. American lungfish)**

Protopterus **(African lungfish)**

***Neoceratodus* (Australian lung fish)**

Coelacanths

- arose in Devonion period
- related to lungfishes and tetrapods
- *Latimeria chalumnae* and *Latimeria menadoensis* – in are the only two known living coelacanth species
- Fins - have eight fins : 2 dorsal fins, 2 pectoral fins, 2 pelvic fins, 1 anal fin and 1 caudal fin.
  - three-lobed caudal fin- diphycercal tail
- Scales – cosmoid scales that act as armor
- Notochord- retain an oil-filled notochord
- Lung - fat-filled single-lobed vestigial lung, homologous to fish swim bladder.
- Kidney - two kidneys, fused into one, located ventrally within the abdominal cavity, posterior to the cloaca.


Coelacanth
Latimeria chalumnae
Sail-like first Dorsal fin
Second Dorsal fin
Bulbous 3-lobed tail
Bony scales covering body
Black eye
Teeth only in front of mouth
Gill cover
Fleshy pectoral fin
Ventral fin
Anal fin
©EnchantedLearning.com

Class Actinopterygii

- More than 23 600 species
- Earliest actinopterygians $\longrightarrow$ Palaconiscids

Characteristics

- Skeleton - ossified
- Gills - single gill opening covered by operculum
- Fins - paired fins supported by dermal rays

- Limb musculature within body
- Teeth with enamaloid covering
- A swim bladder may present
- Palaconiscids gave rise to two groups

1) Subclass - Chrondrostei

paddle fishes, bichirs and sturgeons

2) Subclass – Neopterygii

Gars, bowfin and teleosts

Subclass - Chondrostei

Characteristics

- Heterocercal caudal fin
- Skin covered by Ganoid scales which are rhomboidal in shape
- Usually present a spiracle
- More fin rays than ray supports
- eg. paddle fishes, bichirs and sturgeons

Paddle fish - *Polyodon*

sturgeons - *Acipenser*

Bichirs - eg. *Polypterus*

- Freshwater
- have lungs and other primitive characters resembles to paleoniscid ancestors
- In shallow water, *Polypterus* inhales primarily through its spiracle

Subclass - Neopterygii

- Originated during Mesozoic era

Characteristics

- Skeleton - primarily bone
- Caudal fin - usually homocercal
- Scales - cycloid, ctenoid scales present
- Fins – paired and median fins present
  - rays support the fins

Two genera of early neopterygians

Amia (Bowfin)

Lepisosteus (Gars)

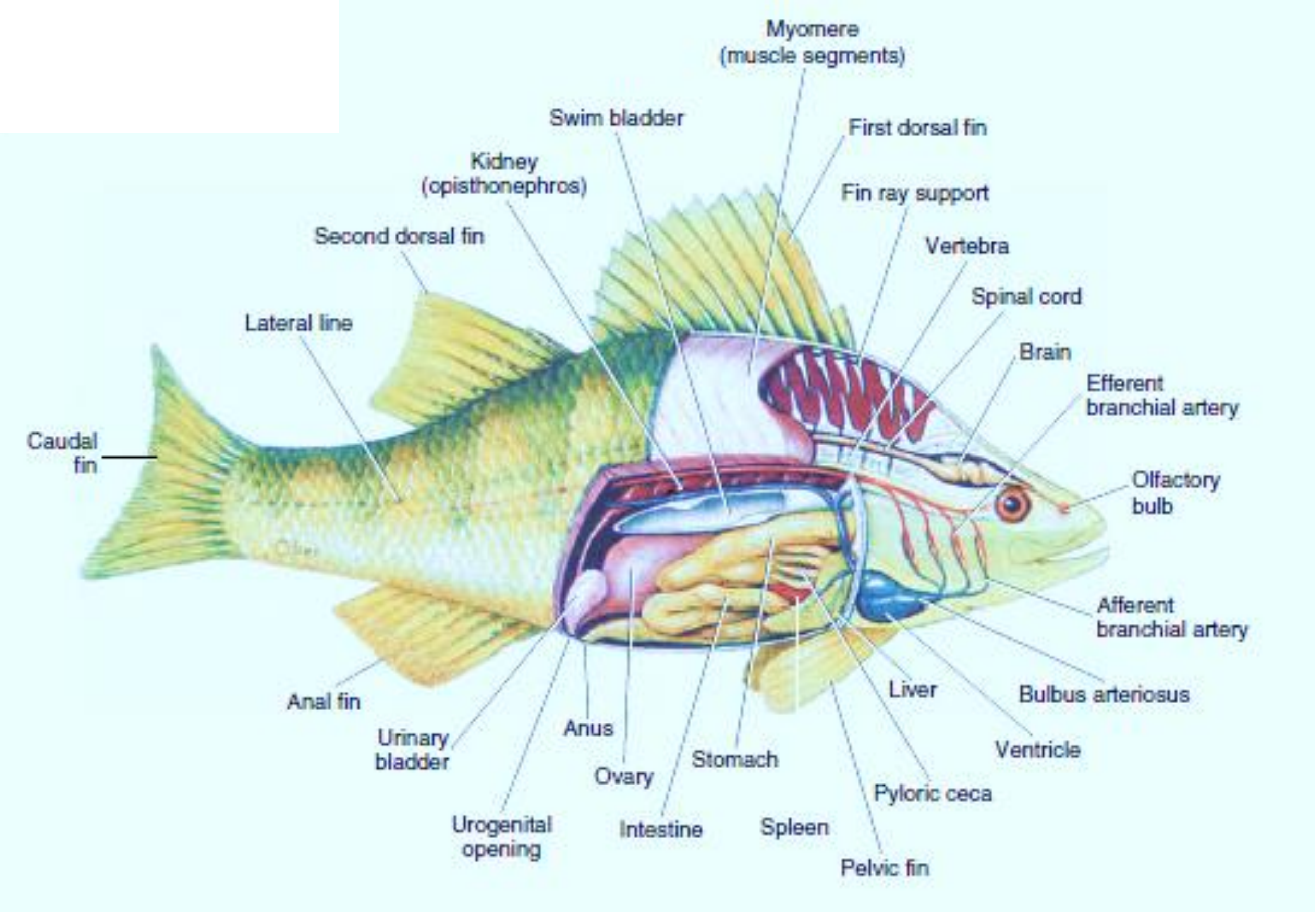
- Gars are large ambush predators with needle like teeth on jaws

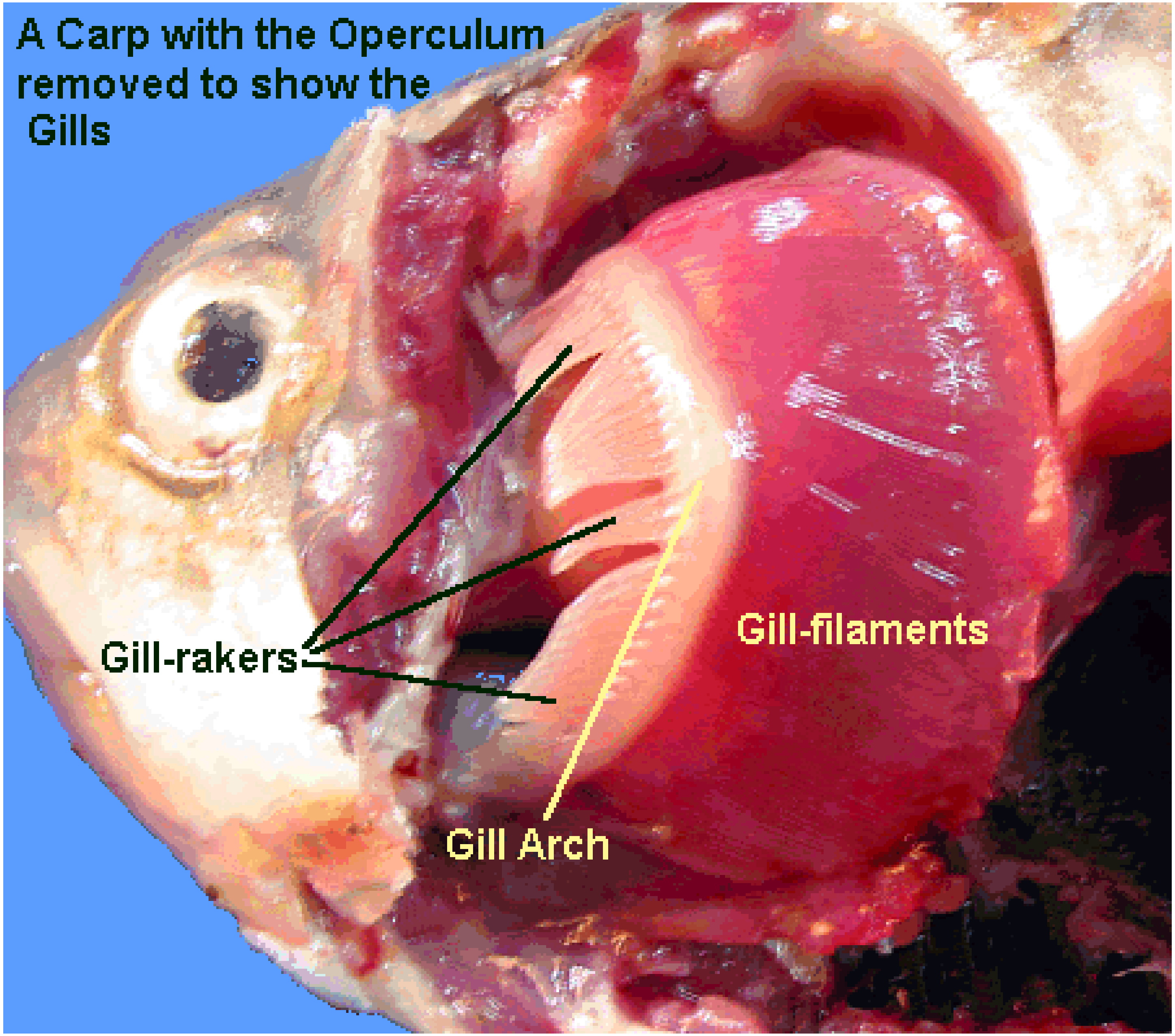
- Major linage of neopterygians → teleosts the modern bony fishes

***Amia* (Bowfin)**

***Lepisosteus* (Gars)**

Internal anatomy of bony fish




A Carp with the Operculum removed to show the Gills
Gill-rakers
Gill-filaments
Gill Arch

The course on FISH BIOLOGY in your second year will give a detail study of fish !!!

Evolution from water to land

- Lobe fins are rare among living fish and are only possessed by the coelacanth and lungfish.
- However, lobe *limbs* are possessed by many living organisms — including humans.
- Therefore all tetrapods, share a more recent common ancestor with the coelacanth and lungfish than with ray-finned fishes.

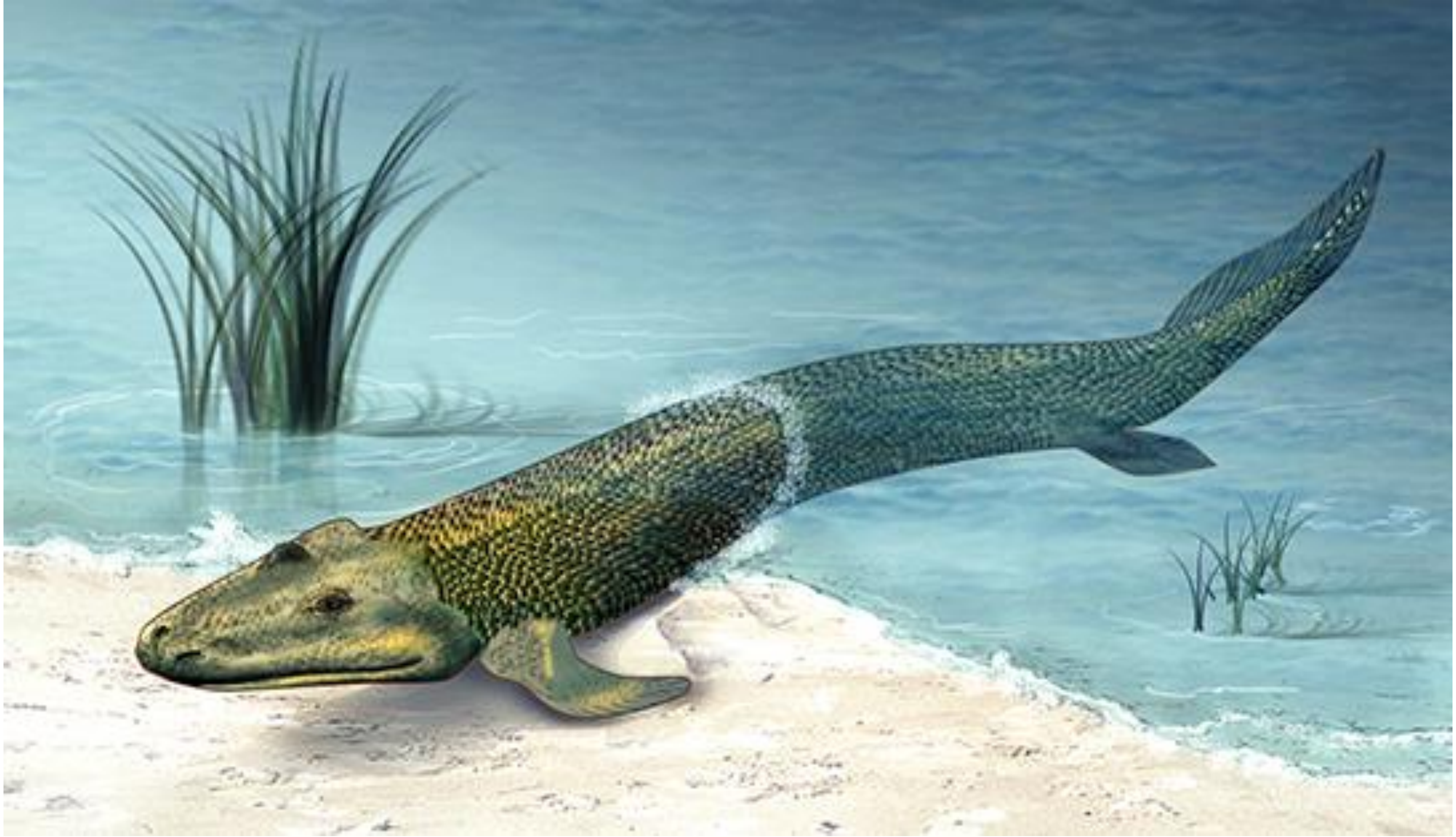
- the first tetrapods were amphibians and they evolved around 395 million years ago during the Devonian period from lobe-finned fish – sarcopterygians (coelacanth and lungfish)

- lobe-finned fish have a bony base that allowed the subsequent evolution of the limbs of the first amphibians.

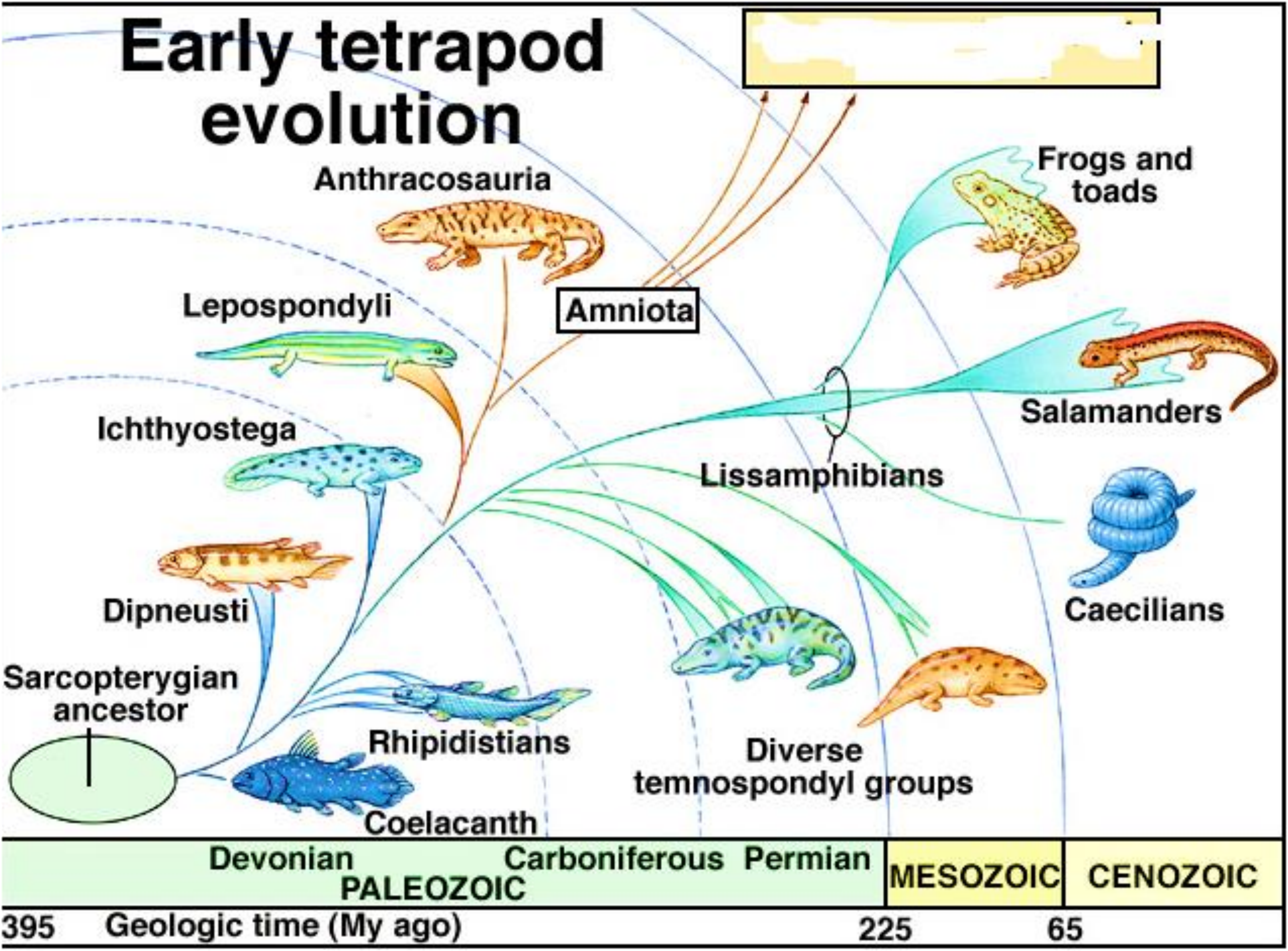
- within the sarcopterygians, the nearest relatives of the tetrapods arebelong to

  order - Osteolepiformes - a group of tetrapodomorph

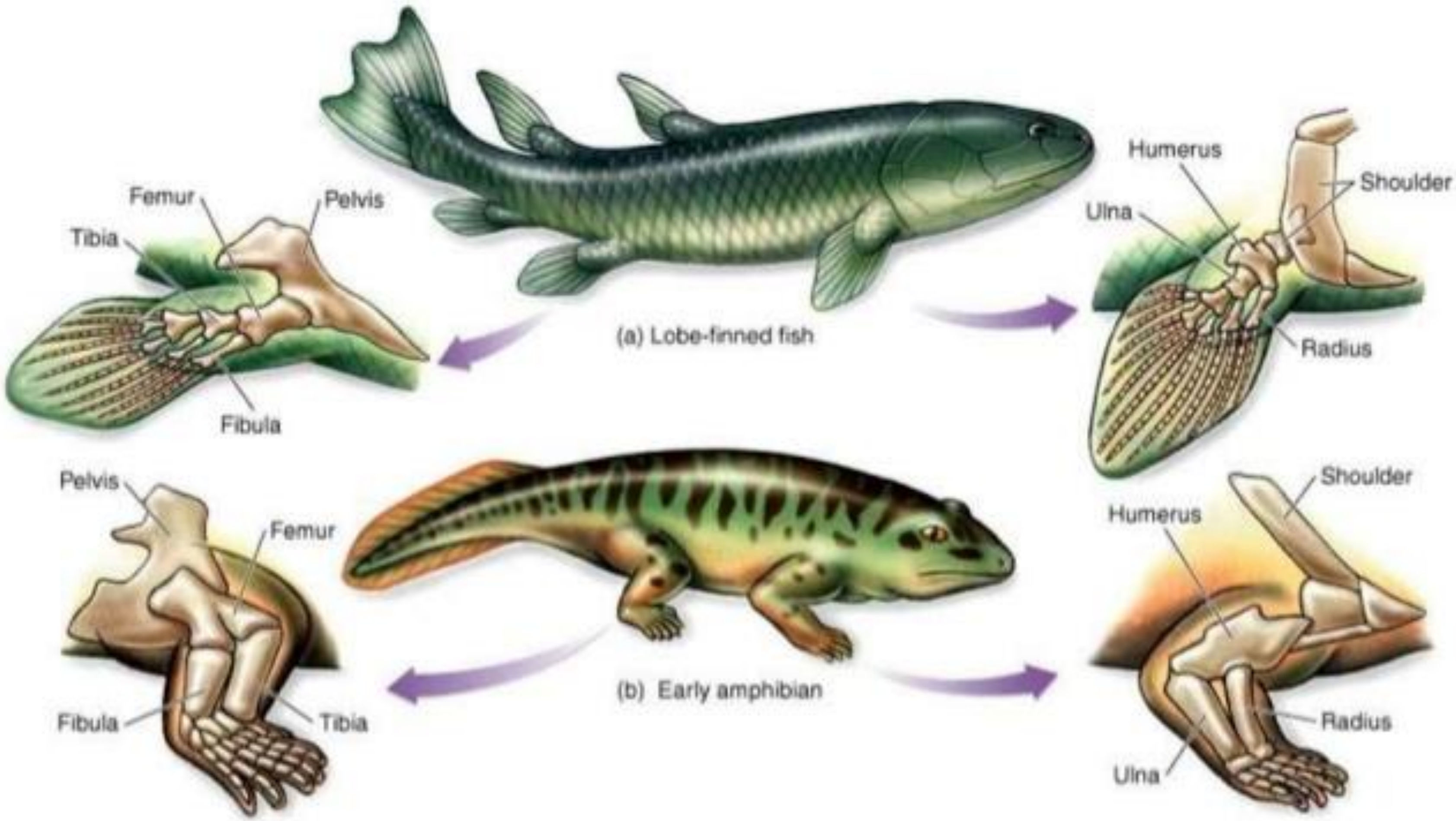
  fish that got extinct about 299 million years ago.

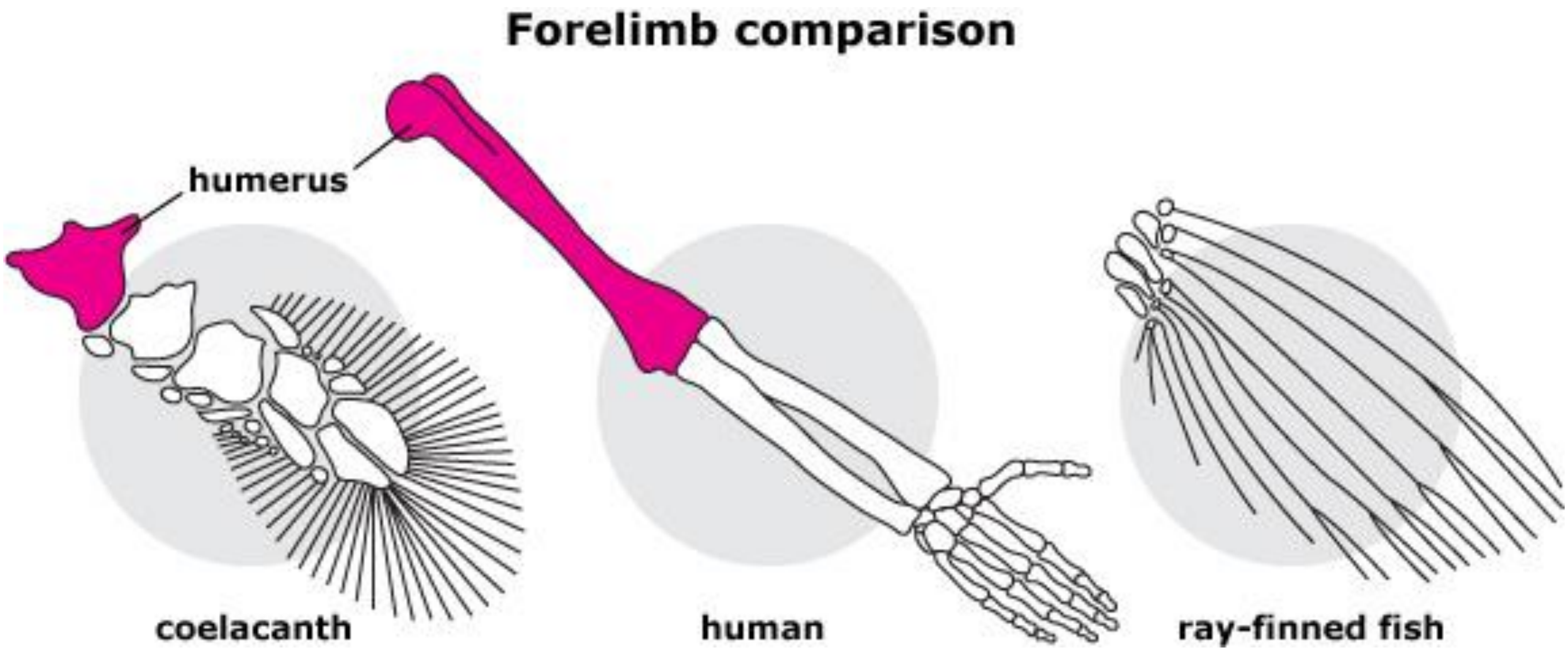


Tiktaalik roseae: The fossil discovery of *Tiktaalik roseae* suggests evidence for an animal intermediate to finned fish and legged tetrapods


Early tetrapod evolution
Anthracosauria
Frogs and toads
Lepospondyli
Amniota
Salamanders
Ichthyostega
Lissamphibians
Caecilians
Dipneusti
Sarcopterygian ancestor
Rhipidistians
Diverse temnospondyl groups
Coelacanth
Devonian
Carboniferous
Permian
PALEOZOIC
MESOZOIC
CENOZOIC
395
Geologic time (My ago)
225
65

Lobed-finned fish vs. Amphibian Bones




Forelimb comparison
humerus
coelacanth
human
ray-finned fish


The END